

G10 E-Portfolio

What is globalization and interconnectedness and its significance in space and time?

Globalization is the closer integration of the countries and peoples of the world, which has been brought about by the enormous reduction of costs of transportation and communication. Including the breakdown of artificial barriers to allow the flow of goods, services, capital, knowledge, and people across borders. In summary, globalization is the idea of connecting countries and people together to allow the flow of goods and services from and to different countries in order to be interconnected and dependent on one another. Interconnectedness is the state of having different parts and or things connected or related to each other. In space, globalization has led to the blossoming of global economies where markets extend beyond national boundaries. Companies like BMW, H&M etc. can operate internationally, sourcing materials labor and customers from various locations. Cultural exchange rapidly occurs, shaping and spreading global values, identities, and norms. Over time, globalization has enhanced historical processes by condensing distances and timelines. Trends and innovations that are in a part of the world can frequently and rapidly impact others, these include topics like politics, economics, sciences, and social dynamics. The significance of globalization lies in its potential to promote global and local growth, or more specifically economic growth, technological innovation and cultural understanding. However, it also raises concerns regarding inequality, environmental degradation and cultural homogenization. Balancing and understanding the complexities of globalization are crucial for addressing global challenges in both the present and future in order to foster a sustainable future.

To what extent has the impact of games shaped, developed and brought communities together in space and time?

The impact of games on shaping, developing and bringing communities together in space and time is extensive and extensive. Thought to be conceived as nothing but recreational activities, games have expanded into powerful tools for social interaction, education, development of skills, cultural exchange and more. Games have bypassed geographical boundaries, enabling people from various corners of the vast world to connect and commune in virtual environments. Online multiplayer games, gaming centric social media platforms and esports events have contributed to this global connection. The connection of fostering friendships, alliances, and even rivalries between players. On top of that, games stimulate collaboration and teamwork, nurturing communication, problem-solving skills and empathy within a vast range of diverse communities. Games have influenced cultural norms, technological advance, and even social dynamics, starting trends and having an effect on economic growth, becoming integral parts of the global popular culture.

However, concerns that need to be addressed are associated with addiction, toxicity in online communities and the portrayal of violence and preconceived notions persist. Nevertheless, the overall impact of games and online communities shaping, developing and uniting various communities across space and time is undeniable. Offering the possibilities for beneficial social change in parallel with challenges that call for careful stewardship.

Strand A1: Explain and justify the need for a solution to a problem for a specified client or target audience.

Client

- Who is your client?

My client is a student in 10th grade who enjoys playing a wide range of video games with and without friends. My client's favorite game is Super Mario bros as well as games like rooftop snipers. My client has a very close friend that visits my client's house often. My client only has a computer and plays 2 player games whenever the friend visits. My client prefers aesthetics in games that are bitlike but have a wide variety of themes, palettes, and colors. My clients provide a large emphasis on a choice of theme and large accessibility that allows my client to adjust settings and personalize their experience.

Problem Statement

- What problem(s) is your client facing in relation to the project theme? Explain the problems.

My client owns an iMac and wishes to play two player games with their friend on the same pc, unfortunately there are not a large variety of games that fit their preferences. They prefer platformer games and competitive games with a large emphasis on a variety of choices and themes. Furthermore, my client enjoys playing games with other players on the same keyboard. Games such as rooftop sniper, basketball jump and other 2d platformer games is what she finds playing the most both at school and at home with their older sister

- What techniques is the client currently using for the issue?

Currently, my client is entwined in the cycle of playing stagnated games, lacking unique updates or updates at all. The situation leads to repetitive gameplay caused by the evident lack of fresh content and unique added features. Thus, dulls the gaming experience as well as stalling and hindering the sense of progress and discovery, which is one of the most important senses for easy, fast and consistent dopamine release. The stagnation of the 2 player input market not only affects my client but also their friends and people they play with, sharing the fatigue, demotivation, and deprivation of enthusiasm. The repetitive nature of the games blurs collective enthusiasm, leading to a lingering sense of disenchantment. Both my client and their friends yearn for a unique game experience consisting of various different themes, creating new challenges and experiences aimed to rejuvenate their passion for 2 player input games.

- Why is the client facing these problems?

Due to the scarcity of developers, studios and creators contributing to the growth and expansion of this gaming experience. The evident lack of dedication in the creation of 2-player local games has brought about a considerable absence of originality and unique aspects within the industry. Although there are an abundance of similar games featuring tropes within the industry focusing on mono-scaled themes and gameplay mechanics. This level of uniformity originates from the limited assortment of active developers innovating in common niches, leading to a repetitive landscape in which games often both feel and look indistinguishable from each other. Thus, the evident absence of unique experiences and various approaches hampers the evolution of 2 player local gaming experiences.

- What does the client want from you?

My client wishes for me to develop a 2 player local multiplayer game that can be played on a single keyboard that offers a range of options, visuals and a unique experience. The product should contain a number of visuals instead of text, text should not be longer than 5 words per line. Furthermore, to invoke a unique experience, I have to implement design features to make the game replayable and will make the game different from other 2 player input games.

Target Audience

- Who is it a problem for ? Describe your target audience.

My aim for my product is to engage adolescents, children, and adults who enjoy playing multiplayer games, particularly on PCs. To specify my audience further they would have to enjoy 2 player games with a single input, my focus is creating a unique experience that is able to successfully captivate and retain such a diverse audience. In turn the fostering of emotions such as excitement, competitiveness and camaraderie through gaming adventures whether it being a competitive match or cooperative mission the aim is to provide the player with a thrilling, inclusive and unforgettable experience that brings players together regardless of age.

- Why is it a problem for them?

The gaming industry faces a crucial shortage of 2 player games, these games compounded by inconsistent updates and common clichés. The scarcity results are a rapid upsurge in player disengagement, driven by apathy, limited user choice and a lack of unique experiences. With a lack of diverse options and ongoing updates, players quickly grow bored, craving the fresh experiences and meaningful gameplay

interactions that cannot be found. It's imperative to developers and studios to address and identify this gap. Nonetheless, the priorities of game developers to expand into other trending titles restricts and diminishes the amount of updates and games released annually.

Need for a solution

- What would happen if the problem is not solved? How is the absence of this product affecting the client, target audience and possible others?
- If the shortage of 2-player games persists even with a lack of means of addressing it, several adverse effects may become apparent. Primarily the target audience, comprising adolescents, children, and adults who enjoy multiplayer games on PCs will continue to experience the boredom and frustration as addressed previously leading to a decline in the communities size and vibrancy. Moreover, the absence of 2 player games can impact the social interactions between players by hindering opportunities for shared experience and for the younger audiences the opportunity to develop communication and problem-solving skills.

Strand A2: Identify and prioritize primary and secondary research needed to develop a solution to the problem.

Research Area (What will I research)	How will I gather this information - Primary/ Secondary sources	Priority and reason for priority (1- most imp)	Why will I research this information?
How to animate sprites?	Secondary source - YouTube	1	I lack the knowledge to animate a sprite by utilizing different images to show complexity in my game. To make my game booth look good ad properly function need to make sure the animations match the movement and actions
How to adjust volume in singular scenes and multiple?	Secondary source - YouTube or forum	4	I will implement custom music to my game, managing the volume of it is an accessibility made sure that the user is satisfied. Also, the fact that music should be able to be managed throughout multiple scenes makes it easier for the user.
How to code Blinking text?	Secondary source - Forum	5	To maximize the gaming aesthetic, a blinking or flashing text full of instruction or text showing instructions is a classic. Knowing how to do this maximizes the nostalgia of gaming and brings back old gaming tropes.
How to code projectiles?	Secondary source - YouTube	2	Projectiles is an essential element in my game, my game involves a 2 player platformer game and is about 2 players playing a game on 1 keyboard battling each other. To secure a victory one player must defeat the other using projectiles therefore learning how to code the speed and trigger as well as max range is important.
How to code health bars by hits?	Secondary source - YouTube	3	Similarly with projectiles a health system not health bars in particular are essential, to play the game you need a health system and projectiles to determine the winner.

Strand A3: Analyze a range of existing products that inspire a solution to the problem.

Inspiration 1



Source:

Fire boy and water girl

<https://www.coolmathgames.com/0-fireboy-and-water-girl-in-the-forest-temple>

Analysis

Aesthetics -

- Art style

The game has a set theme, although there are multiple games, each one has a set theme such as a temple or a forest. The art style is simple but effective, firstly the sprites are very simple looking, as well as their animations however the animations and transition to the animations are what make them special, it is slick and smooth. Furthermore, in terms of the platform and background, it is detailed enough to help the player understand where the game is currently set but does not overcomplicate it.

- 'Feel of the game'

The 'feel' of this game is a very upbringing and energetic atmosphere, the music allows for a relaxed gaming session focusing on cooperation rather than competitiveness. Furthermore, the aesthetics focus less on the quality of such and more on the game by emphasizing the importance of the journey rather than what is seen by adding obstacles and other mechanisms such as levers and coordination practices to improve teamwork.

- Color usage/pallet

The usage of colors differs according to the theme, however the colors used are both simple and complex, the simplicity in color of acid, lava, and water makes it look simple whereas the aesthetics of the background make the game stand out. The contrast of the bright and dark colors from magma, water, and acid to the rocky and dark jungle temple and passageway.

Cost -

In terms of cost, the game is completely free, it can be accessed by one large number of websites such as friv.com, poki.com and more without a cost. The game does not feature ways to progress faster, you have to play the game to reach the end instead of buying something or skip[ping to the end.

Customer -

The target audience for this game are the younger ages, specifically those around or below 8 years old. The game itself focuses on cooperation rather than competitiveness and helps young people to develop skills with communication, team building, chemistry building and cooperation skills. The types of customers that would enjoy playing this type of game would as previously stated be the younger ages, the simplicity of the game stands out, the controls are simple, and the game is user-friendly.

Environment -

- The Goods

The goods of this app environmentally are that while the game may not directly address environmental concerns, they can serve as a platform for raising awareness and educating players about the environmental challenges that lay ahead. Game developers can incorporate themes of sustainability, conservation, or environmental advocacy into the game content or mechanics to promote sustainable behavior among players. Finally, the large player base and community of this game can serve as a platform to advocate and inform the dangers that are impacting the planet and environment aimed at the players, especially the younger ones.

- The Bad

In contrast, there are numerous bad things that outweigh the goods, however I wish to highlight one, energy consumption. Playing fire boy and water girl requires electronic devices such as computers or mobile devices. These devices consume primarily electric energy. Whilst the energy usage of playing a singular game can be small, it can be significant, as the cumulative effect of millions of players worldwide contribute to the overall energy consumption and carbon emissions.

Size -

The size of the game can fit mobile, computer, and TV devices as an all rounded game, in terms of size the game Fire boy and water girl: elements weighs at 177mb of storage. Each level results in around 2 to 5 minutes on average, with a total of tens of different levels resulting in around 30 to 90 minutes to finish the game at minimum.

Safety -

In terms of safety, the game does not require any safety badge or meet any requirement, it is suitable for all ages and does not involve violence or any actions that are deemed dangerous.

Function -

The primary function of the game is to provide players with engaging problem-solving gameplay experiences that ties in with cooperation. The game challenges players to think critically, plan their moves and coordinate actions between the two players to progress to the increasingly difficult levels. The purpose is to entertain and stimulate problem-solving skills, as well as the promotion of cooperation and teamwork in multiplayer games.

Material -

- Made from html5,CSS and JavaScript, making it compatible with many of the most modern web browsers and platforms such as laptops, tablets, and phones. Only if they have a compatible web browser, downloading external and extra software is not required.
- Fire boy can interact with the fire elements and collect red gems, while Water girl can manipulate water and collect blue gems. Players must use these abilities strategically to navigate through levels, avoid hazards, and reach the exit door with all the gems to progress to the next level. The game progressively introduces new elements and mechanics, such as teleporters, switches, and movable objects, to increase the complexity of puzzles and keep players engaged.

What are the strengths of this existing product?

1. Function

The idea of aiming to create a puzzle like game that advocates the importance of cooperation instead of competitiveness sets this game apart from the rest. Furthermore, the fact that levels are progressively increasing in difficulty is a key strength.

2. Aesthetics

The simplicity of the sprites and designs that are bright stand out, as well as the contrasting colors of red and blue contrast with the darker and more subtle colors in the background. The use of simple sprites and animations but refining them to make it more appealing is a key strength.

What are the weaknesses of this existing product?

1. Highly repetitive

After playing the game once a large amount of the players won't repeat the levels unless they have to and they probably won't know or play the next game in the series days after initially playing the game.

1. Simplicity

Although the simple background is good in terms of color contrast, it lacks creativity as the background does not change and it keeps its 2D image which is not as appealing as a 3d detailed background.

What ideas can you take from this existing product?

- Have a somewhat detailed background that can help the player(s) depict the location or setting of the game, but not over-thinker or overdone as to confuse players and disrupt their attention whilst also retaining simplicity.
- Allow the user to choose how they plan to do an action in order to win, allow multiple options and make the user(s) think critically and plan ahead as well as adapt to situations.

Inspiration 2

Screenshots of product



Source:

(author, title, year of publication, page number or web address)

Super Smash Bros. for Nintendo 3DS and Wii

Analysis

Aesthetics -

The art style of the game is unique, it is vibrant, detailed and even has some unique esthetical designs and features like the smash bar and sash effects. Furthermore, the game allows a wide range of diverse maps, differentiating in backgrounds, platforms and sprites. The user interface is sleek and intuitive, providing players with essential information without cluttering the screen. Menu designs are clean and well-organized, enhancing the overall user experience.

Character movements are smooth and responsive, making battles feel dynamic and exhilarating. The animations seamlessly transition between different actions, ensuring that gameplay remains fast-paced and engaging. Super Smash Bros. offers a wide roster of characters, each with unique play styles and abilities. This diversity ensures that every match feels fresh and offers new strategic possibilities, contributing to the game's lasting appeal. Many stages feature interactive elements and hazards that keep matches unpredictable and exciting. These stage hazards add an extra layer of challenge and strategy, encouraging players to adapt their tactics on the fly.

Cost -

The cost of the game is £38.90 and includes some in game purchases for DLCs for added characters. To run the game you require either a Wii or 3Ds however these in-game purchases don't benefit the player except for a larger variety of characters to play.

Customer -

The target audience is aimed at a wide range of audiences from children above 8 to 12 to even adults as it offers a competitive atmosphere as well as an assortment of characters from famous and long lasting games and movies.

Environment -

Similarly to the previous inspiration, the game does not have any features that claim to provide positive change or impact on the environment. The use of the game in terms of charging the console and playing online accumulates electricity from players worldwide, which in turn cumulates overtime to result in a large use of fossil fuels being used, thus polluting the atmosphere with excess and large amounts of carbon dioxide.

Size -

The size is 2.5Gb and can only be downloaded and played on the Wii or 3Ds.

Safety -

The game consists of using physical violence consisting of kicks and punches in addition to forcing the opponent off the platform with powers, making the game a PG12.

Function -

The function of the game is to provide a sense of happiness and create dopamine from the fun characteristics of the game, making this game for entertainment but also a competitive environment for players who wish to improve and show their skills. The game features multiple playable characters that allow for a diverse range of choices and play styles, thus allowing players to learn a vast range of mechanics and skills. Which in turn allows for exciting competitive gameplay featuring many people broadcasting their skills and strategic tactics.

Material -

The use of C++ and C# is what comprises and makes this game possible. However this game can only be played on the 3DS and the Wii.

What are the strengths of this existing product?

- **Fast paced combat**

The use of fast movement and a small map allows for fast encounters, and long matches that make the player use various techniques and plan out before playing, allowing skill curves and learning curves to form. In addition the vast range of characters allows for unique techniques, tactics and gameplay mechanics to be used inducing users to play with various play styles such as ranged, up close or hybrid tactics creating unique gameplay and allowing the player to prove their adapting and quick thinking skills.

- **Vibrant visuals**

The vibrant visuals stand out and make the game look appealing, it details the setting and provides an atmosphere, either upbeat, pressure and such.

What are the weaknesses of this existing product?

- **Map diversity**

Although the background diversity and map themes are large, the platforms itself dont present a diverse layout usulting in the use of repetitive strategies being in continuous use.

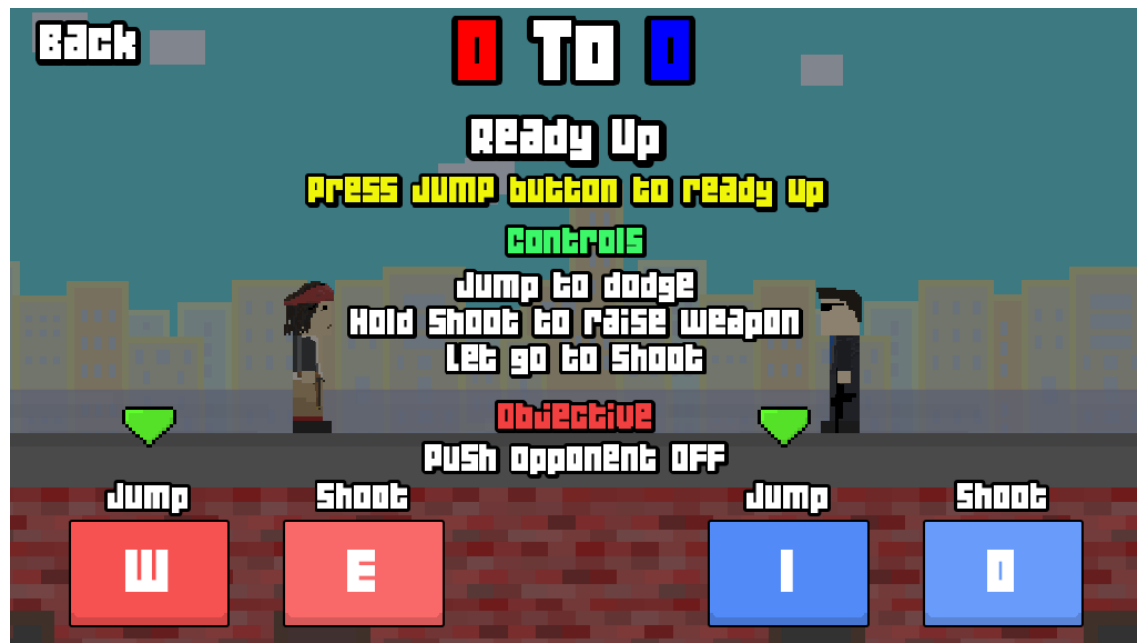
What ideas can you take from this existing product?

- Fast paced movement

The use of fast paced movement in a small ish map to allow fast encounters and prolonging matches as well as gameplay mechanics that players can learn and master.

Inspiration 3

Screenshots of product



Source:

(author, title, year of publication, page number or web address)

<https://poki.com/en/g/rooftop-snipers>

Analysis

Aesthetics -

- Art Style

The game has a set theme, the theme is about 2 individuals on a rooftop trying to shoot each other off. This game offers a variety of different sprites that the player can choose. The game combines simple cartoon graphics with detailed sprites, the picture above shows a simple design of a rooftop and the simple background of a city. However, if we turn our attention to the sprites, they are in comparison heavily detailed.

- ‘Feel’ of the game

The ‘feel’ of the game is unique, the controls are simple to understand in combination with the cheerful and upbeat music, however in contrast to this the game is very different. The game itself is an on the edge of your seat game, providing a thrilling and fast-paced experience that pressures and rewards the player to play fast in order to win.

- Colors

The colors used in the game are simple yet effective, primarily you have the red and blue indicating who is player one and who is player two. The information shown results in colored bright text to indicate the topic of the text in combination with a white text bordered by black outline. This is especially effective due to the darkening of the background, allowing the brighter colors to show their vibrancy.

Cost -

The game is completely free with no in game purchases and makes the user play the game to progress without any shortcuts.

Customer -

The target audience for this game are the younger ages, consisting of younger adolescents and below till the age of around 5. There is violence in the game so it may have a negative effect on the audiences that are too young, furthermore the game is simple and does not focus on anything else but gameplay for the most part.

Environment -

This game is not eco-friendly, although the energy usage whilst playing this game is minimal due to simple graphics and low gameplay mechanics in contrast to other games. A large population plays this game, thus the energy usage accumulates altogether, utilizing more electricity and thus the combustion of more fossil fuels with the aim of supplying for this demand.

Size -

The size of the game can fit solely computer and portable (tablets and phones) devices, in terms of size the game rooftop snipers weigh 45.4 MB of storage. The time it takes per game depends on it, however it can last a couple of seconds to over a minute.

Safety -

The game promotes the use of violence, the aim of the game is to use a sniper or projectile based weapon to shoot your opponent in order to win or beat the game. According to the Apple Store, this game is for those who consist of ages twelve and above. This is true due to usage of guns and minor bits like gore that is detailed.

Function -

The purpose of this game is solely entertainment, by providing a fast-paced and intense game into the market that utilizes simple controls, it allows for a diverse range of audiences to play the user-friendly game. Furthermore, the use of guns and gore may entice others to play this game. To play the game, one must either play it on a suitable compatible web browser or download it to an appropriate device.

Material -

The game is made from the following coding languages: HTML, UNITY web, CSS and JavaScript. Thus allowing the game to be played on most browsers and devices such as windows, mac, Linux, tablets and phones.

What are the strengths of this existing product?

The strengths of this product are

- Aesthetics

The aesthetics are simple and unique, the details on the background are simple yet they provide a good setting to the game only depicting what is needed. In contrast, the details on the player models are details to draw the players attention to it as well as being the only thing customizable by the player it is able to transitions between animations smoothly

- Size and accessibility

The small size allows people not to worry about it taking space, furthermore due to the availability on a large population of platforms it allows people to play on almost whatever whenever.

What are the weaknesses of this existing product?

- Environment

The impact of the environment this game has over accumulative time is concerning. A large player base resides in playing this game, thus meaning that a large amount of energy is used overtime from the large player base.

- Repetitive

The rounds progress quickly, thus making the game lose its taste and initial effect it has on the audience. There are no obstacles to go around and the game does not require skill.

What ideas can you take from this existing product?

Accessibility

Allows the product to be accessible on multiple platforms including a range of computer, laptop and portable devices. Furthermore, allow the game to run offline and from anywhere to make sure the accessibility is almost universal no matter what situation you're in.

1. Background Information

Explain what the project is all about and who it's for.

This project is centered around the idea of addressing a notable issue within the gaming industry, this being the scarcity of games that invoke local competitiveness and communication across various dimensions of space and time. The project specifically targets individuals that seek gaming experiences designed and modified for two players sharing a single input device such as a keyboard. The aim of this project is to create an environment that encourages competitive engagement, the strengthening and building of bonds and effective communication between players. By highlighting the aspects stated the project searches to fill a niche in the gaming industry by catering to players needs and values such as the dynamics of shared gameplay and a sense of community.

2. Objectives and Goals

Explain what you hope to achieve with the project. What problem are you trying to solve, or what message are you trying to communicate?

The aim of my project is to create a game centered around the needs of players looking for games focusing on the dynamics of shared gameplay on a single input device in an arid market. By developing a game in such a diverse industry with the aim of targeting those that seek gaming experiences for two players that contains both competitive engagement and communication. In a time in which people play video games more than going outside the game aims to intertwine both aspects of social interaction and entertainment by providing a two player game that involves a single input device.

3. Deliverables

List all the things you need to create for the project, such as sketches, mockups, or final designs.

To create my project, I require materials to help me plan and evaluate my prototypes. I will use:

- Mockups
 - This app will be able to provide me with the tools to create drawings and diagrams to present sizes and various features to create my final product
- Google forms
 - This allows me to gather information from possible clients and people who have experiences with websites like this and give me productive feedback to help improve the app.
- YouTube
 - This will help me by providing extensive information regarding ways in which I can create a better product, furthermore if stuck whilst coding it can be able to support me.
- Unity forums
 - This will provide me with help from a variety of people who have similar issues and present ways to fix them if required

4. Timeline

Outline when each part of the project is due, so you can stay on track.

What is expected	Date
Crit A	09/02/24

Crit B drawings/diagrams	10/02/24
Crit B	15/02/24
game	22/02/24
Crit C	22/02/24
Crit D	26/02/24

5. Budget

Explain how much money you have to work with and any limitations you need to consider.
I will not work with any money and am not planning to invest a certain amount of money for capital, with this a large amount of limitations are brought up, some include, paywalls in apps such as mockups and unity, reduced access to made assets and such.

6. Target Audience

Describe the people who will be using or seeing your design. What do they like, and what do they need?
Summarizing strand Ai, My target audience are people in the following generation; generation Z, Alpha and millennials that have an interest in bit-like games that invokes competitiveness between two people in a portable and unique way. One unique principle that has to be followed is that the game is a 2 player game that recognizes a single input.

7. Branding Guidelines

Provide any general requirements to adhere to corporate style and image. This may include specific colors, fonts, logos, or other visual elements that need to be included in the design to ensure it aligns with the company's brand identity.

- No text
- 1200 x 700 (pixels)
- Logo has to be based around connectivity and or 2 player connectivity.
- No specific color or amount of colors but has to remain mono (no gradient)

8. Communication

Explain how you'll keep in touch with everyone involved in the project.
My client is someone I am familiar with, thus I do have both their contact information and are interacting with them on a daily basis for both the projects and academic intentions. Thus allowing me to easily contact my client whenever needed.

.Strand B1: Develop design specifications, which clearly states the success criteria for the design of a solution.

Aesthetics	Consider appearance, style, color, shape/form, texture, pattern, finish, layout.9 User-Friendly Layout: The design and construction of an organized, simple and intuitive design is paramount with emphasis and prioritization of visuals such as logos for buttons as well as text readability Modern Style with Essential Elements: Utilize a modern design style retaining minimalism designs in combination of various color palettes when required as well as images or logos. Primary colors may differ however the theme(s) have to be clear and invoke a user selected theme in which the user chooses what theme to go for complemented by accent colors to enhance text visibility and visuals Font Selection for Readability and Style: Opt for non-serif fonts to maintain simplicity and readability. Utilization of logos in buttons: incorporate a wide range of visuals to instruct what the button is for and use a hexagon or circle
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	shape of button for simplicity and familiarity whilst capturing the audience's attention by utilizing both visuals and colors. Diverse themes and pallets for a unique experience: Utilize and design a range of themes the user can choose between, each theme focusing on one or more ideas with an appropriately selected color palette and graphics. In summary, the website's aesthetics are thoughtfully designed to meet user expectations, blending modernity with thematic elements. Clear layouts, strategic color choices, font selections, and button shapes aim to create an appealing, user-friendly, and readable design.
Function	What will your program do? What are the different features? Competitive: Create and design a 2 player game that encourages a healthy competitive environment among one another in a standoff between skill and arcade fun. Health system: Create and design a health system to determine a winner offering chances and opportunities for comebacks and a unique experience Repetitive: Create and design 3 or more viable maps that have different themes or ways that make them unique
Cost	Is there a maximum cost? Is this a material cost/time cost/selling cost? None except for energy bills and wifi bills. No, the game will be completely free to those who wish to play, the assets and game composition will be made using free means such as free pixel drawers to create assets, import assets and make
Customer	Who is it for? What is the target user's age, gender, socio-economic background? The client is aimed towards young adults and teenagers or more specifically those in the generations of millennium, gen z and gen alpha to experience a 2 player game that invokes socialization and communication that has bitlike aesthetics.
Environmental considerations	Where will the solution be used? How will the design directly or indirectly affect the environment? The solution will be used online, it will impact the environment in a negative way, to use and pay this game you require a platform to play it on an internet to download, the energy required can be either renewable energy or fossil fuels. In addition to maintaining servers and the domain of the game will also require a larger amount of energy.
Development	What resources are available? What is needed for the development of the solution? How much time is needed to create the design? I have a computer and internet as my resource, I plan to design and create with my computer and use the internet and peers to answer any questions have regarding coding and developing. My client can answer any question I have and give me feedback and opinions. I need 1–2 hours to develop designs and list containing the information that will be required
Technology	What will the client require to run it? The client will require internet as well as a web browser to access it, it will be available on every web browser possible to ensure that everyone can access and can download the game from the website, and played on mac, windows and even linux.
Quality	Will all the errors be trapped? Will appropriate messages be displayed for the user? No, because of the app's simplistic design a simple reboot of the game or a change of scenes will fix most problems.

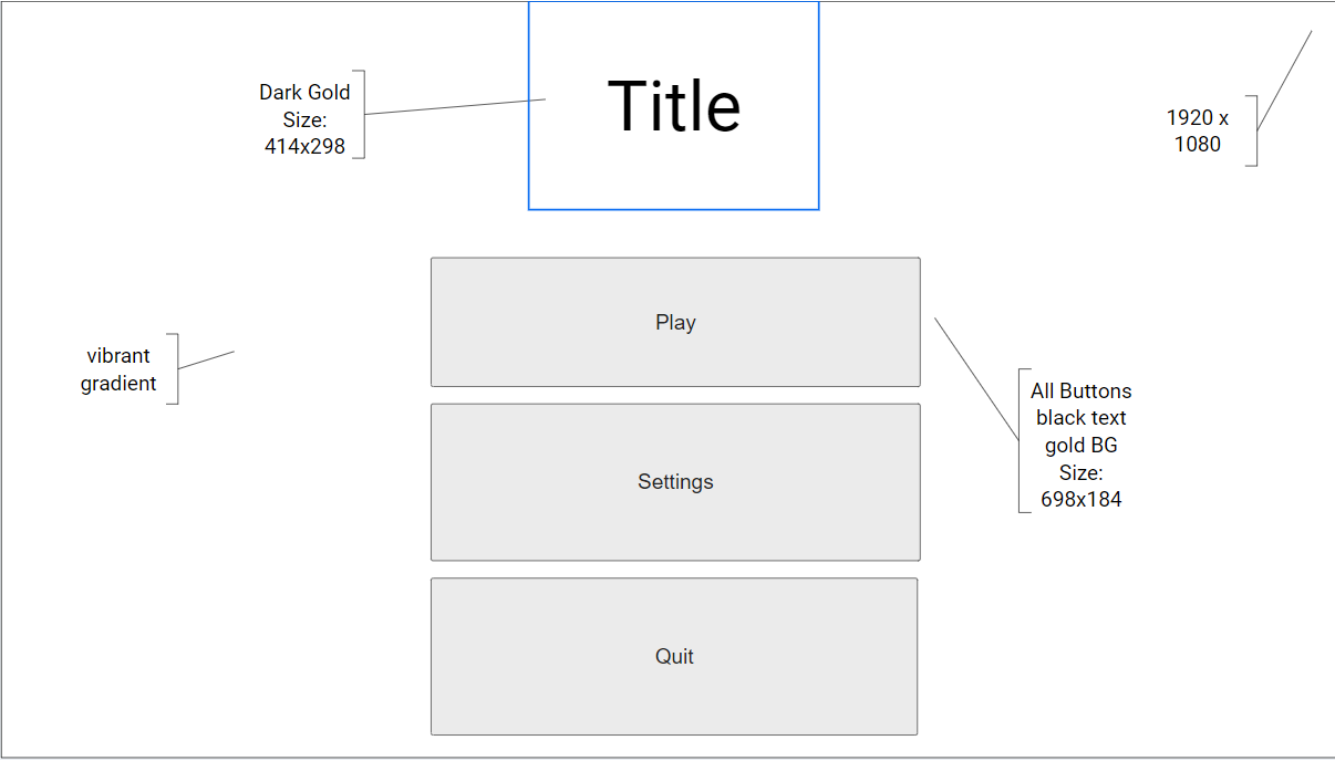
Safety	What safety factors need to be incorporated into the design? N/A
Size	Are there any specific sizes that need to be considered? What “human factors” need to be considered? The app should be able to cover the entire screen or in a minimized view. The sprites used have to be a minimum of 1% of the screen to be able to be recognized and the buttons that should be interacted with should not be too big or too small resulting in a minimal size of 4% of the screen. Finally the main text should be a maximum of 8% of the screen.

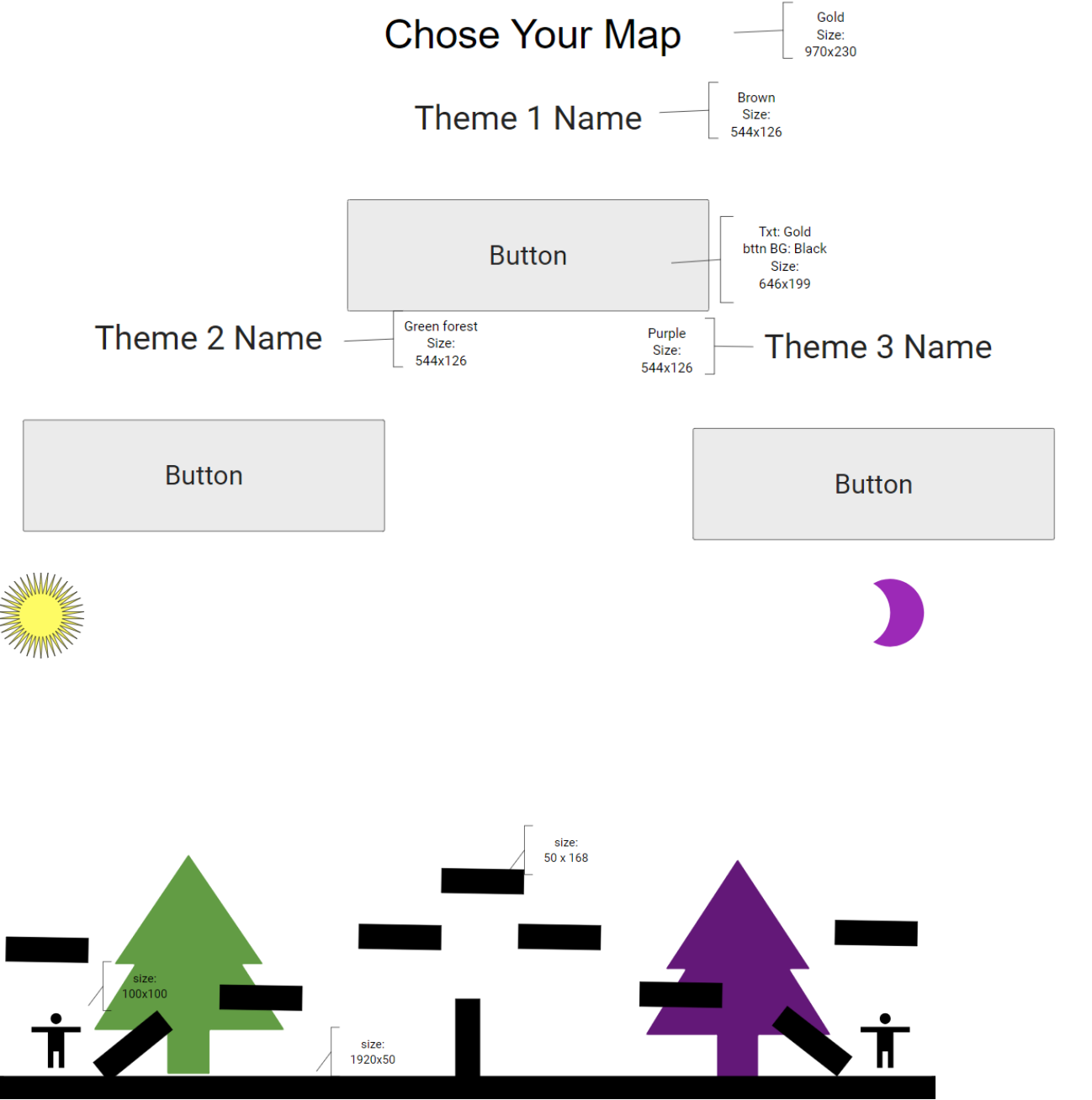
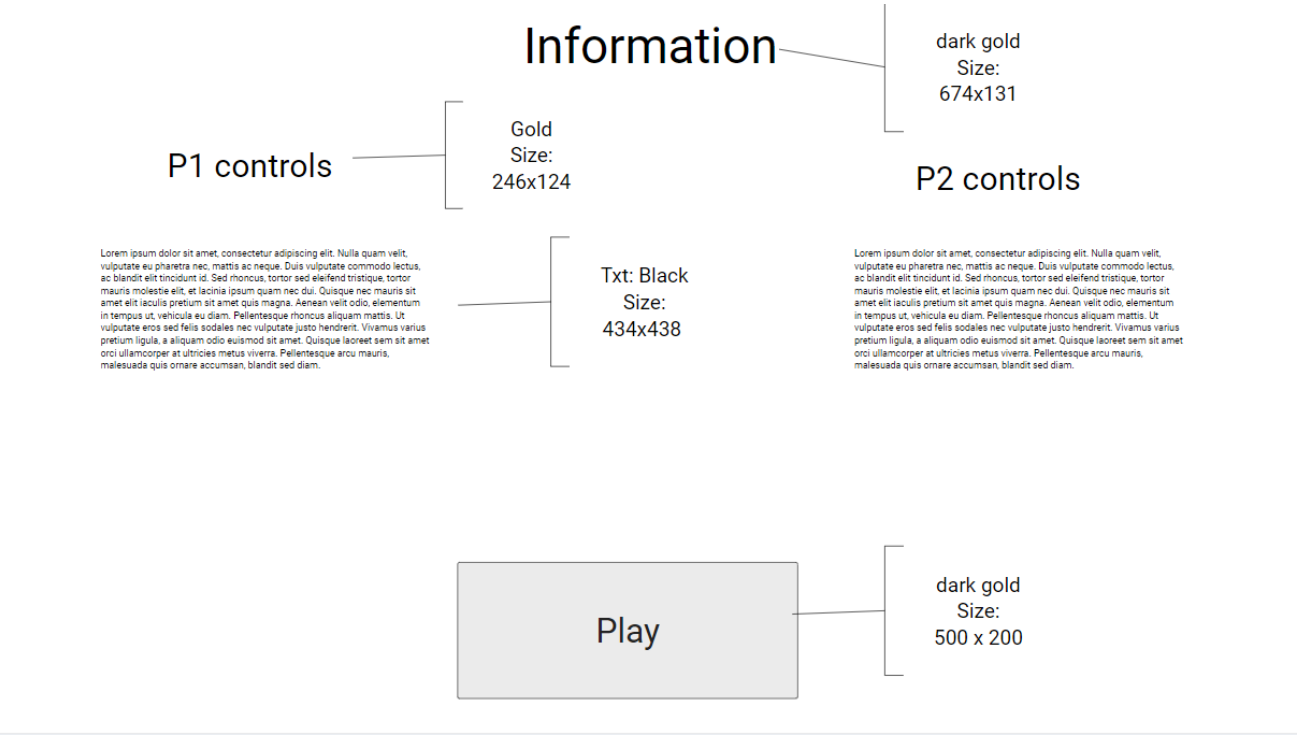
Strand B2: Develop a range of feasible design ideas, which can be correctly interpreted by others.

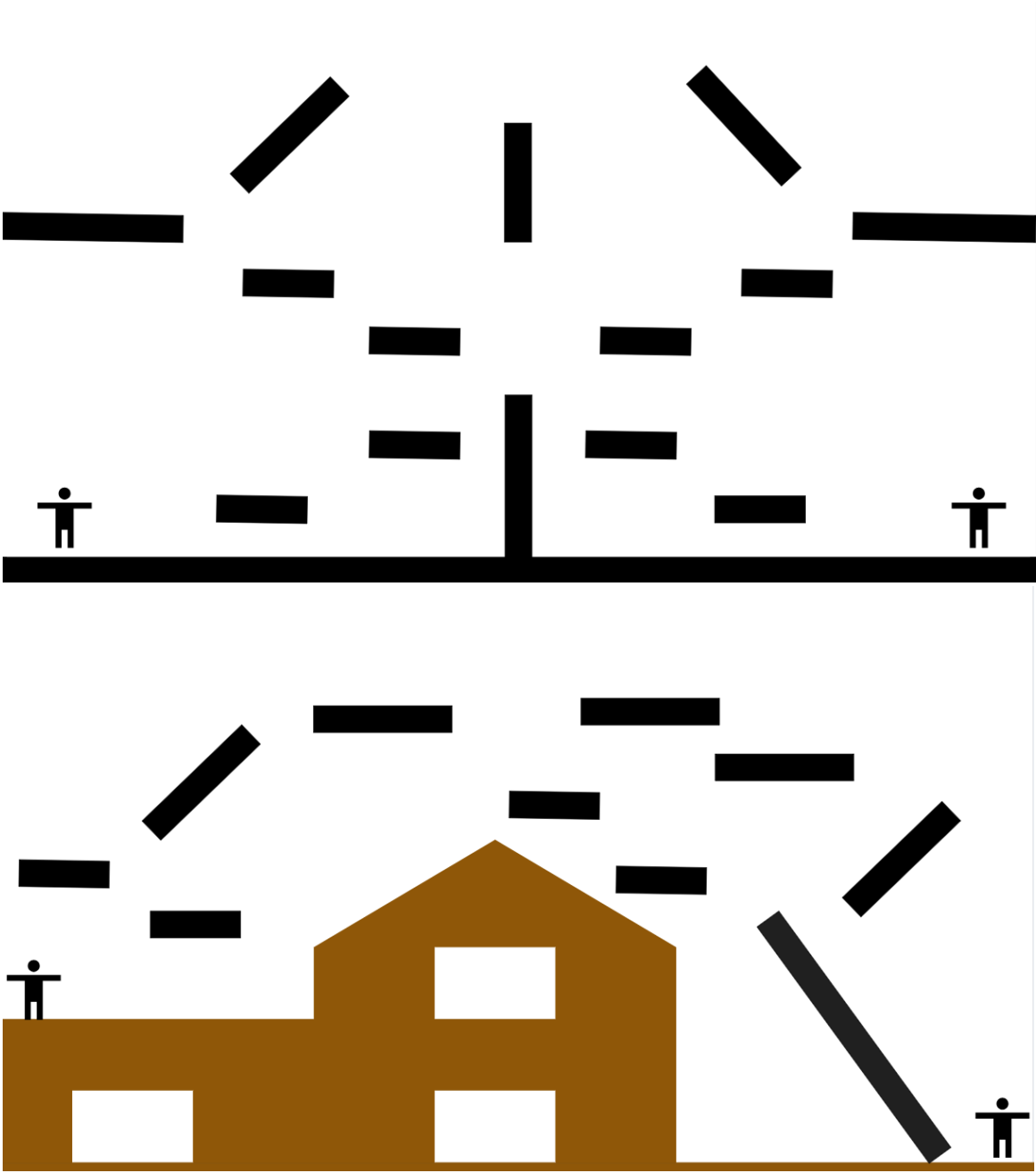
Note:All sized documented are in Pixels or px and are in format: Title page, information page and map selection page
map designs 1, 2, 3 are the same due to them being set as theme placeholders and not exact maps

Design:

1







2

vibrant
gradient

Dark Gold
Size:
414x298

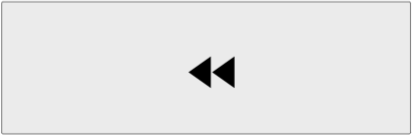
Title

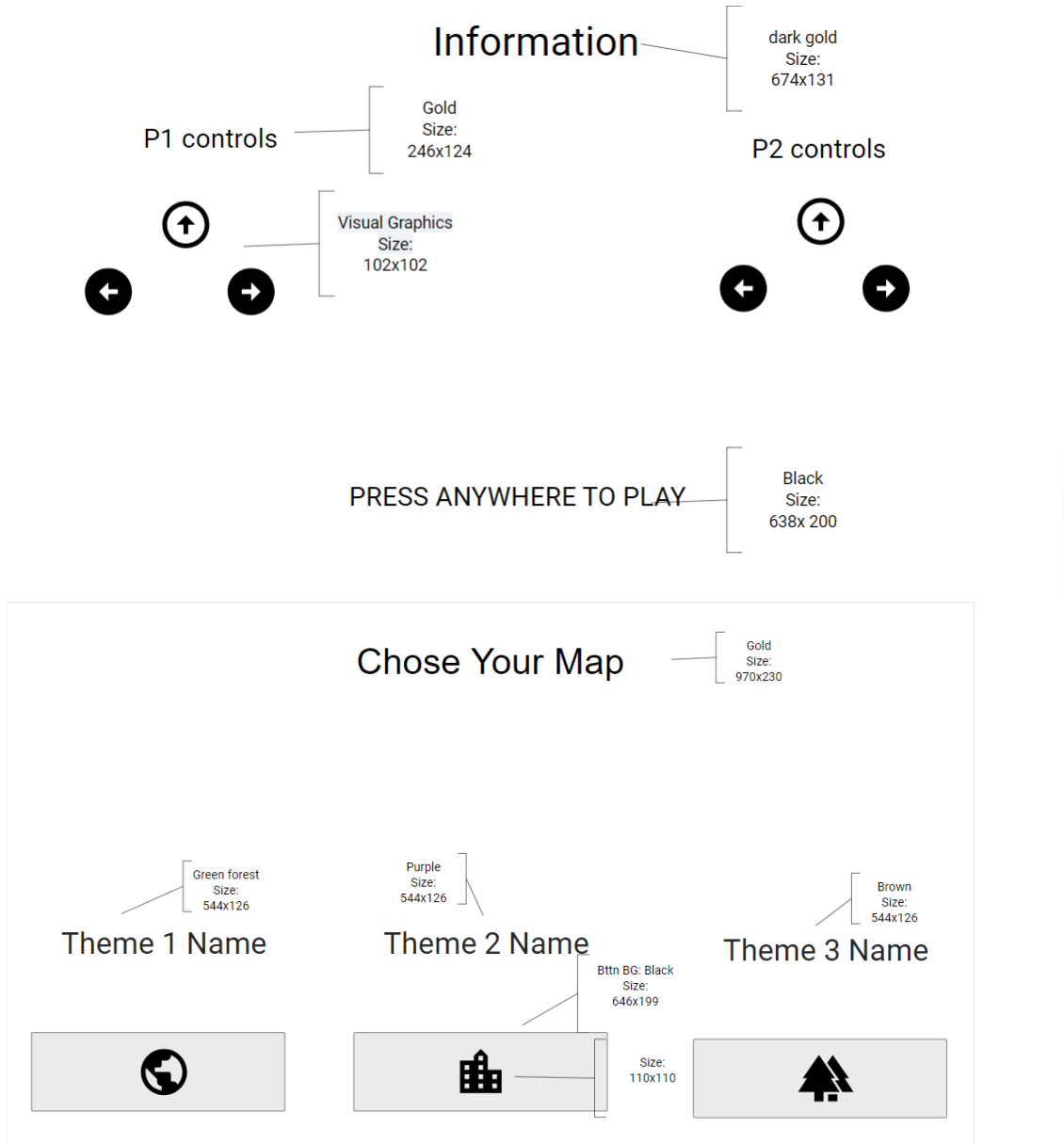
1920 x
1080

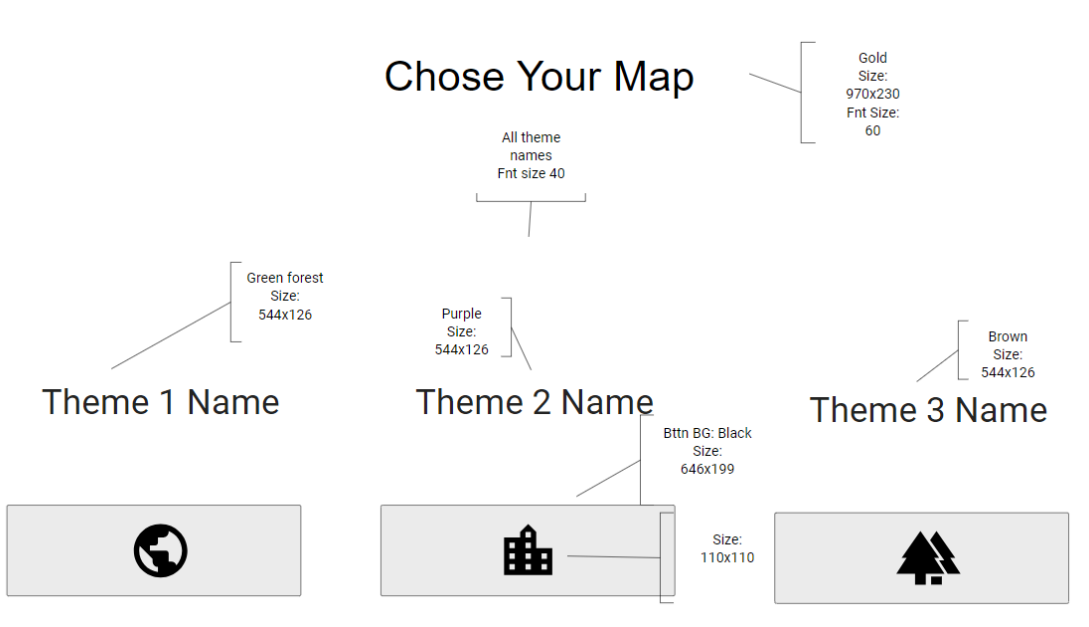
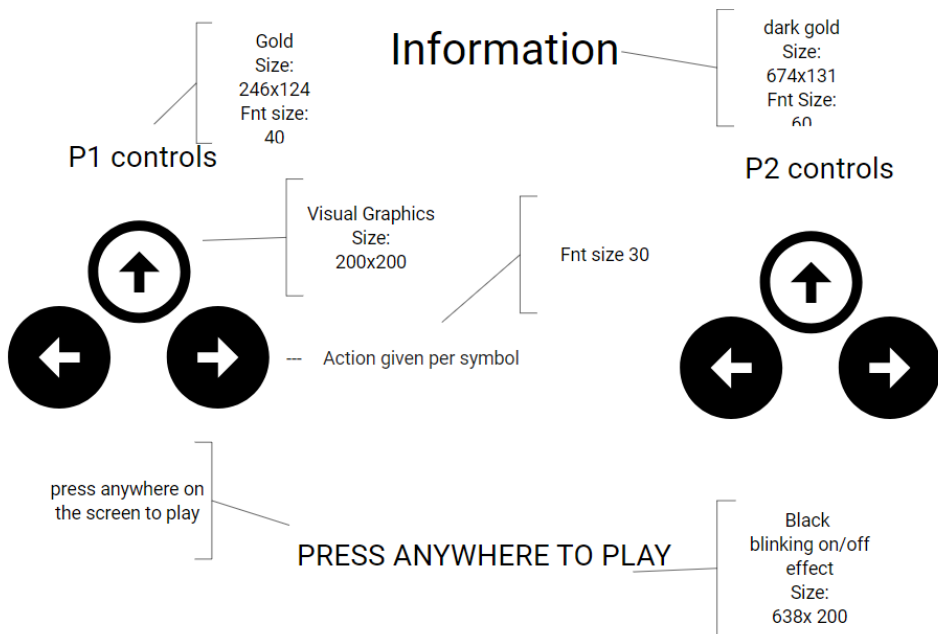
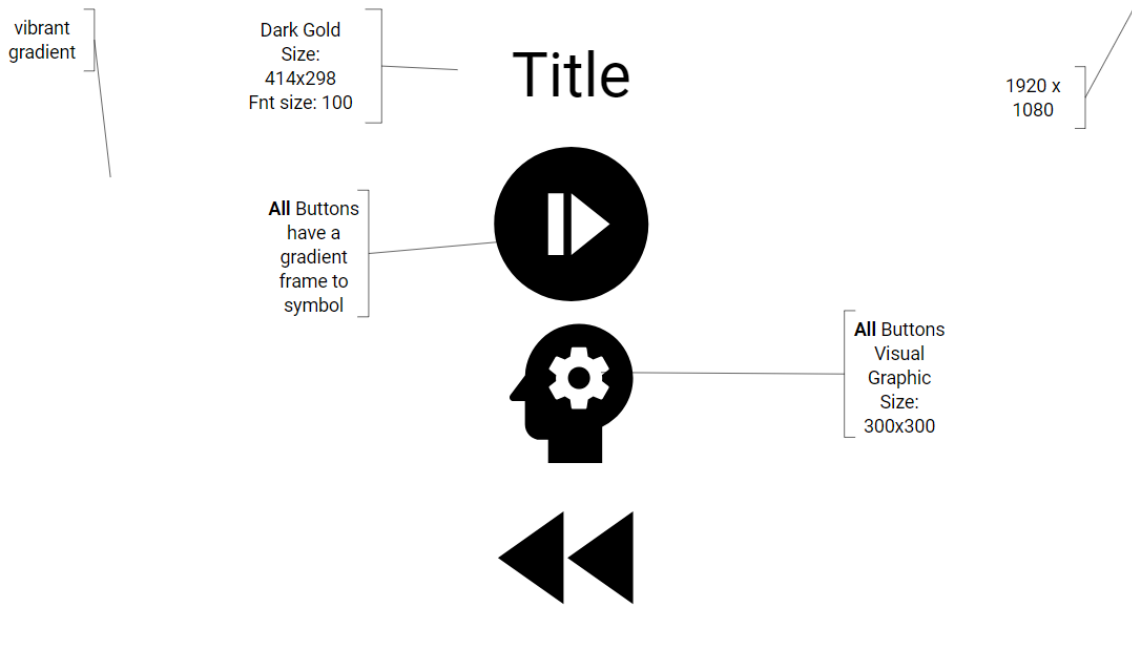
All Buttons
Size:
698x184



All Buttons
Visual
Graphic
Size:
107x107







OR

Refer to Appendix D

Main Body Font Options

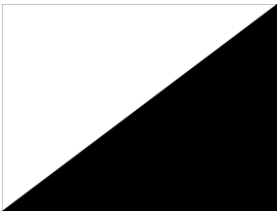
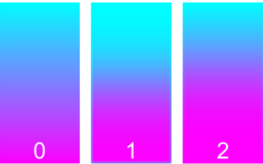
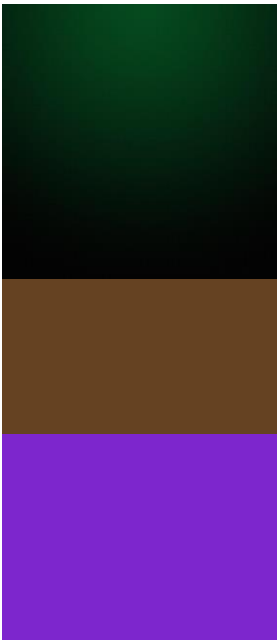
Font Name: Arial	
What it looks like:	Arial aaaaa ABCDEFGHIJKLMNOPQRSTUVWXYZ abcdefghijklmnopqrstuvwxyz ABCDEFGHIJKLMNOPQRSTUVWXYZ abcdefghijklmnopqrstuvwxyz 0123456789 0123456789
Strengths: Versatile simple easy to read Recognised and is familiar by others Compatible with a large amount of devices	
Weaknesses: Overused common	

Title Font Options

Font Name:Roboto	
What it looks like:	Roboto SUNGLASSES Self-driving robot ice cream truck Fudgesicles only 25¢ ICE CREAM Marshmallows & almonds #9876543210
Strengths: Eye catching, distinctive unique	
Weaknesses: Hard to read in large amounts Looks bad in small text	

Colour Palette

Option 1:	Description:			
	Main text color	Background Color	Title Color	Subheading Text Colour

Fill the cell each cell with a color —→				
	#000000 and #FFFFFF	#D512D5 to #12CFD5 via #0E4AD3	#D8AD0D	Bottom to top #7509A1 #A16309 #0A6A10
Strengths:	Main text color ● Simple ● Stand out ● Easy to read ● Not too bold or complex Background color ● Popping vibrancy ● Energetic ● Enthusiastic ● Contrasts with other colors and helps them stand out Title color ● Bold color ● Stands out in contrast to the BG ● mono Subheading color ● Deep dark colors ● Contrast with BG ● Stand out ● Bold ● Simple mono In summary the aim of the main text is not to stand out but to support reading and information gathering, white and black contrast is a very simple and used palette because it works, it's not distraction, simple and can convey its message without disrupting my clients' attention. The background color stands out with its vibrancy, its energetic colors stand out and contrasts with other colors to help it stand out. The title color is a simple yet effective choice, although it may be mono it is bold, dull in contrast to the background gradient balancing out the scheme of things. Finally the subheading colors, similar to the title color, are in the darker shade connecting with the background gradient to stand out more and reflect individual themes.			
Weaknesses:	Main text color ● Very “bland” colors due to frequent use ● simple Background color ● Too bold ● Too vibrant ● May distract attention Title color			

	● A bit bland ● Mono color Subheading color ● Bland mono color ● Bold The overall limitations of the selection of color choices focus on how the colors may be bold and may not match with one another, furthermore the idea of mono color may present itself to be too simple. As well as the background color being too bright.
--	---

Strand B3: Present the chosen design and justify its selection.

CHOSEN DESIGN JUSTIFICATION: DESIGN 3

What is your choice?

Design 3

Why did you not choose Design 1

I decided against choosing the first design options for a numerous number of reasons that prove to be significant. Design 1’s formatting was simple, after reviewing my design specifications, I felt that the other designs meet higher standards. For aesthetics although the design is simple, clear, and easy to navigate. It lacks evident creativity and visual graphics, the lack of visuals do not give it a unique look and experience. For example, in the information page, the information is written in a paragraph format, this may disrupt the user’s experience and the simplicity of the game. In terms of functionality it works, it effectively helps navigate the player through the game, apart from the information page I am concerned about how many will skip it. A large population of gamers prefer not to read the information and instead dive straight to the gameplay.

Why did you not choose Design 2

Unlike the first design, the second design implements icons and textures and other visuals. This allows for easy navigation and directs for the player(s) and makes the product look sharper and professional. However, a key weakness is that like design one the design relies on the contrast between a vibrant gradient and deep and dark mono colors. Furthermore, the icons or visuals in comparison to the buttons are not what would say is aesthetically pleasing. Finally,Ihave identified a large amount of open space, in turn this will create negative feelings, and it will look empty on purpose and not considered a formal product.

Reason for choosing Design 3

The 3rd design is what I believe is the best of both worlds design, it targets the weaknesses of the first two designs and fixes them or improves them. In terms of the Aesthetics implementation, this design utilized the most visuals and maximizes them. For the main menu, instead of buttons, there is an icon or image surrounded by a gradient reflecting the background. Not only that but in the controls' menu I utilized the same idea from the second design and utilized arrows, however this time combined the icons with short one word descriptions to help inform the user of the controls. The improvements in this design are recognizable to many, it simplified many things, being aesthetically pleasing, simple, and functionally appropriate.

Strand B4: Develop accurate and detailed planning drawings/diagrams and outline the requirements for the creation of the chosen solution.

Sizing of Elements
Spacing of Elements

Refer to B2

All size measurements are in px

What tools (software and software libraries) will be used?

No	Requirements	Justification
1	Webflow	I will conduct the creation of my website here,I Will primarily utilize this website for its designing system due to lack of time.
2	YouTube	I need inspiration as well as advice on techniques that can make code simpler by reducing line usage. Furthermore, it can inspire me to add a variety of unique things to the form to make it stand out or be more effective
3	Google (Search Engine)	When I'm stuck or require inspiration, I can always search for methods which can solve my problem.
4	ChatGPT	I will use ChatGPT SOLELY for the purpose of finding missing letters, capitalization errors or missing semicolons or syntaxes, However I will NOT utilize ChatGPT for the purpose of creating code, shortening it or adding code and such.

Strand C1: Construct a logical plan which describes the efficient use of time and resources, sufficient for peers to be able to follow to create the solution.

Plan how you will create your solution.

- Use technical language where possible to describe the task
- What resources are required? (software and software libraries ?)
- How much time does each step genuinely require in minutes?
- Exact date when completing the step

Step	Task	What resources are required?	Time to complete (in mins)	Date of expected completion
1	Detail and construct a mini design brief and specification of what will be constructed	Whiteboard	20	14/11/23
2	Locate and download necessary assets as well as design, draw and create logos or necessary sprites and such	Whiteboard, canvas, unity asset store	60	18/11/23
3	Design main menu and information page scene	Unity, unity asset store	45	20/11/23
4	Design map 1	Unity, unity asset store	45	24/11/23
5	Design map 2	Unity, unity asset store	45	24/11/23
6	Design map 3	Unity, unity asset store	45	24/11/23
7	Code main menu and information page scene	Unity, unity asset store	70	30/11/23
8	Code map 1	unity	20	4/12/23

9	Code map 2	unity	20	4/12/23
10	Code map 3	unity	20	4/12/23
11	Code sprite 1	Unity, unity asset store	60	20/02/24
12	Code sprite 2	Unity, unity asset store	45	20/02/24
13	Design and input music and code pause menu and setting for music	Unity, Garage Band	90	22/02/24
14	animate	Unity, unity asset store	45	24/02/24
15	Final touch ups	Unity, unity asset store	15	26/02/24

Strand C2: Demonstrate excellent technical skills when making the solution.

Some of the technical skills used in the creation of my game:

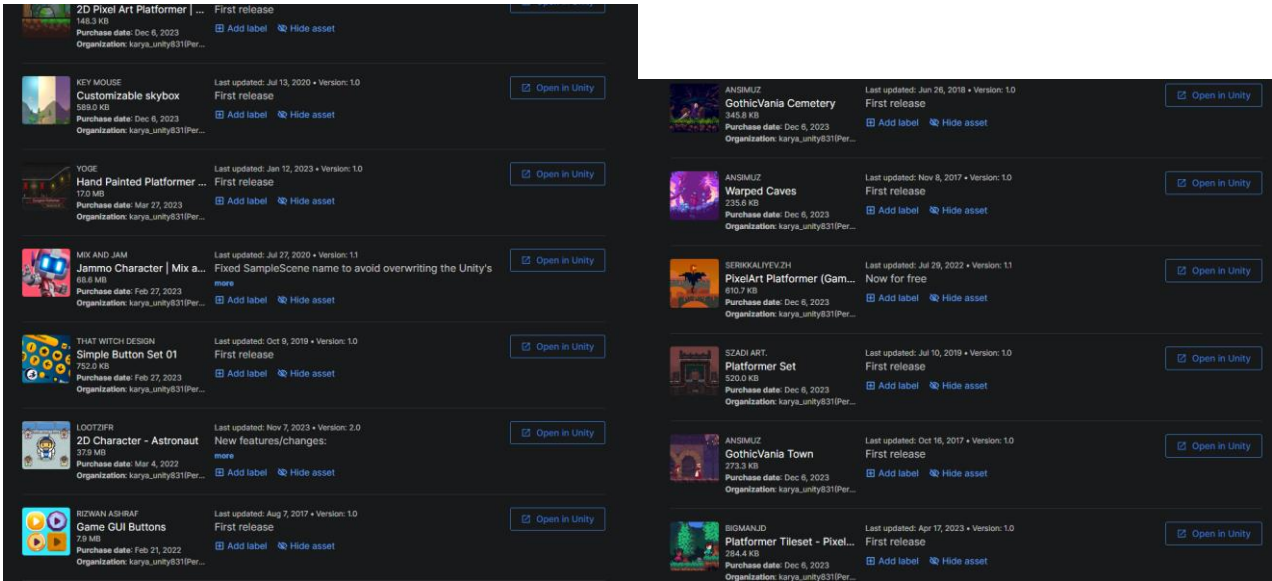
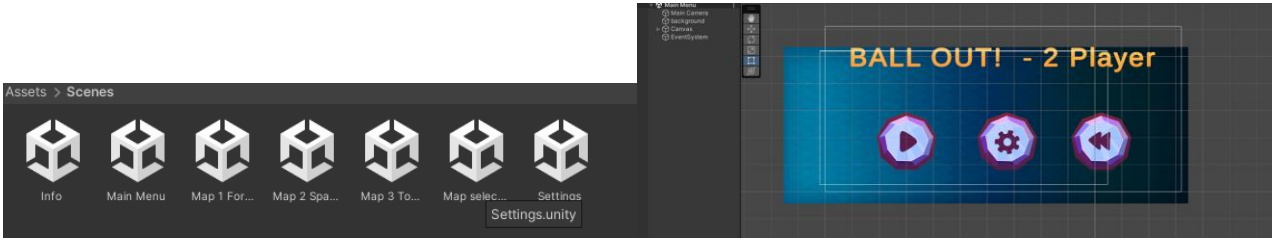
- Variables
- Conditional Statements (if statements)
- Loops (for, while)
- Functions and Methods
- Arrays and Lists
- Classes and Objects
- Unity API
- Coroutines
- Unity Physics
- Unity UI System

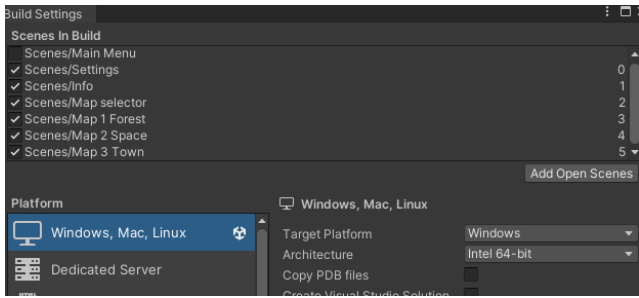
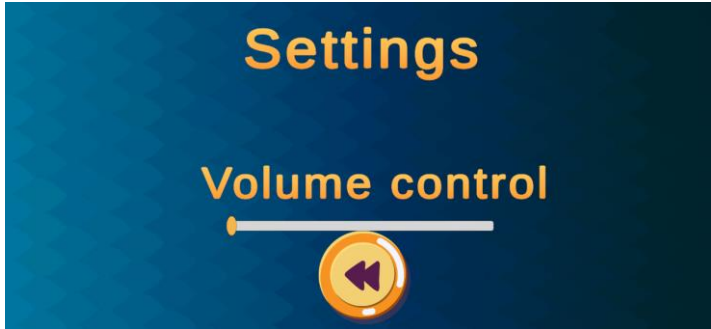
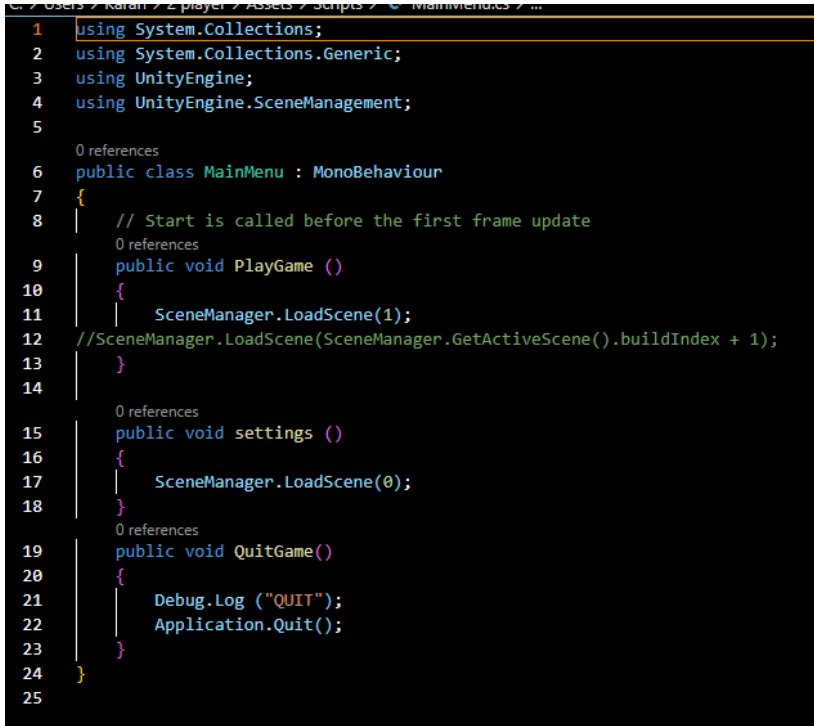
Please refer to C3 for code and Appendix C for the product video

Strand C3: Follow the plan to create the solution, which functions as intended.

Please refer to Appendix C for the product video

Process Journals

Session No.	What happened/ What was done in the session?	
1		This session was about finding and creating appropriate assets to utilize in my game, asIam inexperienced with digital design of assets and the creative side I am experienced with compiling, organizing and utilizing assets. However, I can create simple assets such as 2d hearts and such. AlsoIwas looking around for already existing 2 player games,Iwas experienced in creating 2d platformer games and am confident in it but not 2 player input games. I found games such as fire boy and ice girl helpful as a basis, and rooftop snipers.
2		Today, after brainstorming and reviewing my strand Bii,I Was able to design the main menu and organize all the assets and scenes required.

		
3		This session created a settings menu, inputted a code where on click would go to the information scene. The volume control was from a value of 0 to 1 and adjusted in game music.
4	 <pre> 1 using System.Collections; 2 using System.Collections.Generic; 3 using UnityEngine; 4 using UnityEngine.SceneManagement; 5 6 0 references 7 public class MainMenu : MonoBehaviour 8 { 9 // Start is called before the first frame update 10 0 references 11 public void PlayGame () 12 { 13 SceneManager.LoadScene(1); 14 } 15 //SceneManager.LoadScene(SceneManager.GetActiveScene().buildIndex + 1); 16 17 0 references 18 public void settings () 19 { 20 SceneManager.LoadScene(0); 21 } 22 0 references 23 public void QuitGame() 24 { 25 Debug.Log ("QUIT"); 26 Application.Quit(); 27 } 28 } 29 </pre>	TodayIworked on the main menu,Ihave previously used unity to make 2d games and soIl used most of the code from those games for the menu and interaction. After thisIworked on the setting menu itself and the information menu or controls page. Before coding my load scene program, I ordered them in the build setting and started to look through past games made to utilize the code used in the past. I utilized code from last year and imported scene manager into my script and utilized scene manager to load my different scenes.

```

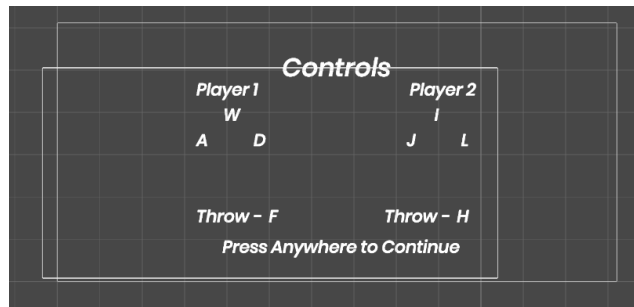
1 using System.Collections;
2 using System.Collections.Generic;
3 using UnityEngine;
4 using UnityEngine.UI;
5
6 0 references
7 public class blinkingText : MonoBehaviour
8 {
9     10 references
10     Text text;
11
12     0 references
13 void Start ()
14 {
15     text = GetComponent<Text>();
16     StartBlinking();
17 }
18
19 0 references
20 IEnumerator Blink()
21 {
22     while (true)
23     {
24         switch (text.color.a.ToString())
25         {
26             case "0":
27                 text.color = new Color(text.color.r, text.color.g, text.color.b, 1);
28                 yield return new WaitForSeconds(0.5f);
29                 break;
30             case "1":
31                 text.color = new Color(text.color.r, text.color.g, text.color.b, 0);
32                 yield return new WaitForSeconds(0.5f); // Fixed typo in WaitForSeconds
33                 break;
34         }
35     }
36 }
37
38 1 reference
39 void StartBlinking()
40 {
41     StopCoroutine ("Blink");
42     StartCoroutine ("Blink");
43 }
44
45 0 references
46 void StopBlinking()
47 {
48     StopCoroutine ("Blink");
49 }
50 }

```

Working on the controls page as difficult,I Decided to code some blinking text due to the game looking bit-like and corresponding to my theme(s).

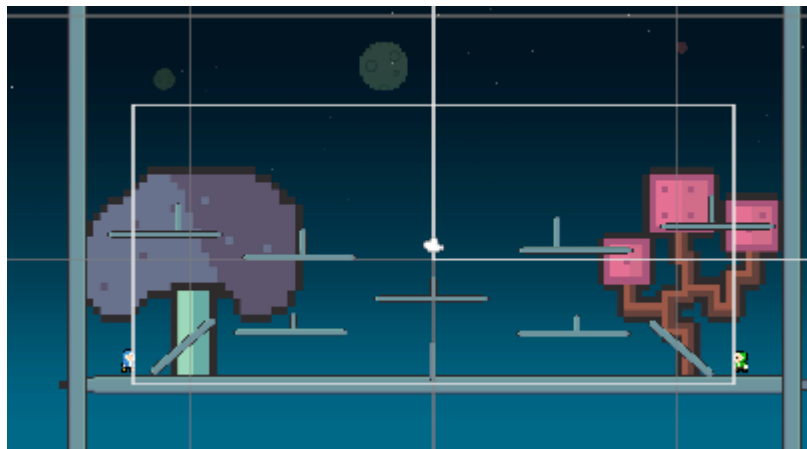
I have never coded in C# a code that affects text, First researched how to make the game identify the difference between text legacy and text mesh pro. After that I identified the variable and created a start program. I had to look at forums by unity to learn about the IEnumerator to make my code work for multiple frames, as well as converting my text color to a string (rgb). From there I learnt about cases which are a selection statement and coded the color changing to white or black and the wait and break timings. Then I used a coroutine (a method that can pause execution and return control to Unity, but then continue where it left off on the following frame) found in unity forums to complete my code.

6



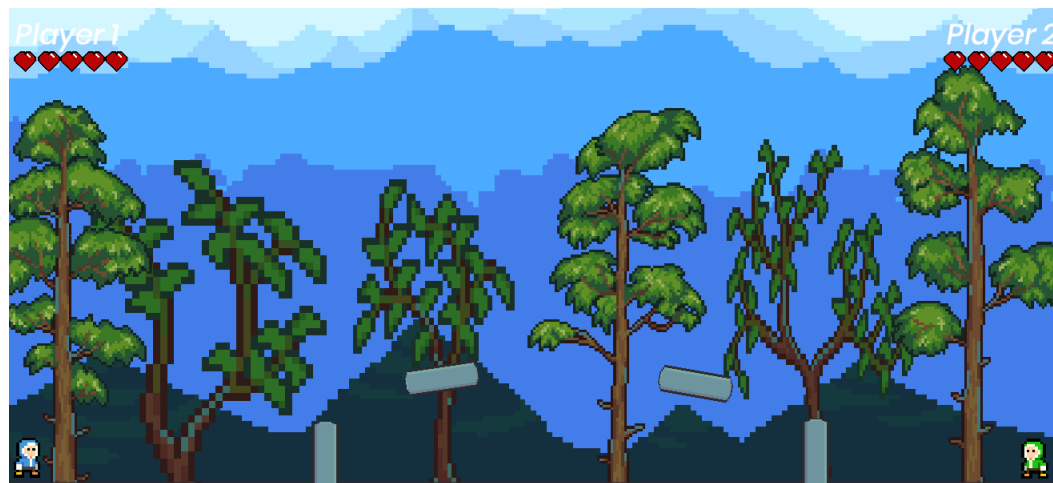
This session made the graphics for controls and applied the blinking text to it. Furthermore, I created the Map selection scene, Utilized load scene manager again for my buttons, text mesh pro for my text and the same background used in the main menu

7



Today started organizing the assets that will use, then I started brainstorming and designing the space map.

8



This session started designing map 1



This session started designing map 3

```

// Update is called once per frame
references
void Update()

    isGrounded = Physics2D.OverlapCircle(groundCheckPoint.position, groundCheckRadius, whatIsGround);
    if (Input.GetKey(left))
    {
        theRB.velocity = new Vector2 (-moveSpeed, theRB.velocity.y);
    } else if (Input.GetKey(right))
    {
        theRB.velocity = new Vector2 (moveSpeed, theRB.velocity.y);
    } else {
        theRB.velocity = new Vector2 (0, theRB.velocity.y);
    }

    if(Input.GetKeyDown(jump) && isGrounded)
    {
        theRB.velocity = new Vector2 (theRB.velocity.x, jumpForce);
    }

    if(Input.GetKeyDown(throwBall))
    {
        GameObject ballClone = (GameObject)Instantiate(snowBall, throwPoint.position, throwPoint.rotation);
        ballClone.transform.localScale = transform.localScale;
        anim.SetTrigger("Throw");
    }

    if(theRB.velocity.x < 0)
    {
        transform.localScale = new Vector3 (-1,1,1);
    } else if(theRB.velocity.x > 0)
    {
        transform.localScale = new Vector3 (1,1,1);
    }

    anim.SetFloat("speed", Mathf.Abs(theRB.velocity.x));
    anim.SetBool("Grounded", isGrounded);

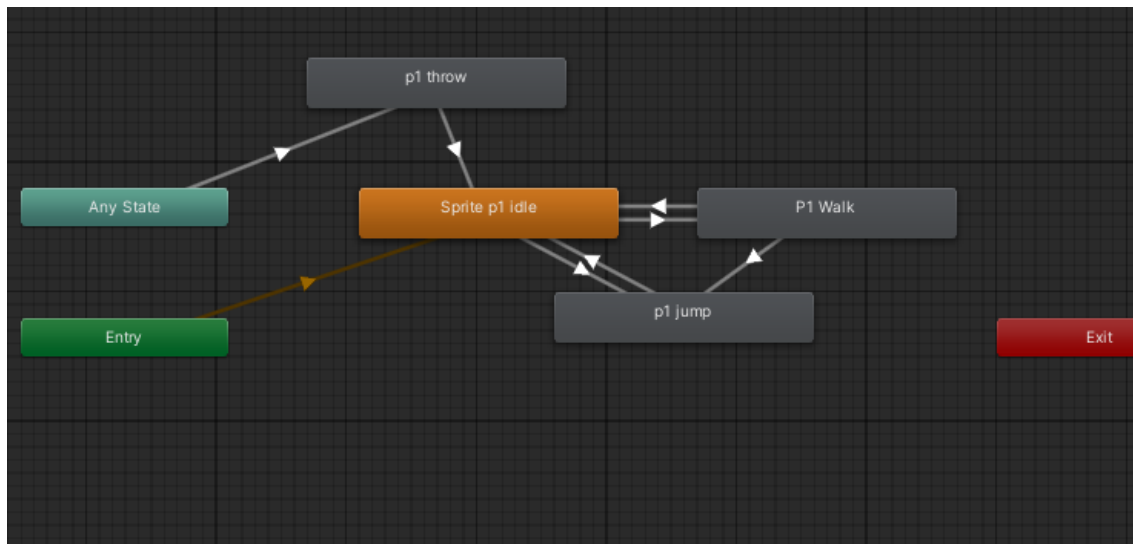
```

Today I started working on player movement. I have previous experience with 2d platformer games in unity so I was familiar with using velocity found from the rigidbody2d and vector 2 looking at the x and y coordinates and used the unity forums to learn new things such as key codes and improved my knowledge of C#.

```

1 using System.Collections;
2 using System.Collections.Generic;
3 using UnityEngine;
4
5 0 references
6 public class Player_Controller : MonoBehaviour
7 {
8     2 references
9     public float moveSpeed;
10    1 reference
11    public float jumpForce;
12
13    1 reference
14    public KeyCode left;
15    1 reference
16    public KeyCode right;
17    1 reference
18    public KeyCode jump;
19    1 reference
20    public KeyCode throwBall;
21    12 references
22    private Rigidbody2D theRB;
23
24    1 reference
25    public Transform groundCheckPoint;
26    1 reference
27    public float groundCheckRadius;
28    1 reference
29    public LayerMask whatIsGround;
30
31    3 references
32    public bool isGrounded;
33
34    4 references
35    private Animator anim;
36
37    1 reference
38    public GameObject snowBall;
39    2 references
40    public Transform throwPoint;
41    // Start is called before the first frame update
42    0 references
43    void Start()
44    {
45        theRB = GetComponent<Rigidbody2D>();
46        anim = GetComponent<Animator>();
47    }

```



Today I did animation, using a YouTube tutorial that broke down how to use unity's animator tool, as well as how to use and change transitions and the difference between different transitions and what makes them effective. Finally, I learned how to change them and what looks good.

```
public GameObject player1;  
1 reference  
public GameObject player2;
```

```
3 references  
public int P1Life;  
3 references  
public int P2Life;
```

```
1 reference  
public GameObject P1Wins;  
1 reference  
public GameObject P2Wins;
```

```
3 references  
public GameObject[] p1H;  
3 references  
public GameObject[] p2H;
```

```
// Start is called before the first frame update
```

```
0 references  
void Start()  
{  
  
}
```

```
// Update is called once per frame
```

```
0 references  
void Update()  
{  
    if (P1Life <= 0)  
    {  
        player1.SetActive(false);  
        P2Wins.SetActive(true);  
    }  
    if (P2Life <= 0)  
    {  
        player2.SetActive(false);  
        P1Wins.SetActive(true);  
    }  
}
```

This lesson, I coded the game manager for health systems. I used a YouTube video and unity forums to create an array, win conditions and heart/player life system

```
public void HurtP1()
{
    P1Life -= 1;

    for (int i = 0; i < p1H.Length; i++)
    {
        if (P1Life > i)
        {
            p1H[i].SetActive(true);
        }
        else
        {
            p1H[i].SetActive(false);
        }
    }
}
```

0 references

```
public void HurtP2()
{
    P2Life -= 1;

    for (int i = 0; i < p2H.Length; i++)
    {
        if (P2Life > i)
        {
            p2H[i].SetActive(true);
        }
        else
        {
            p2H[i].SetActive(false);
        }
    }
}
```

```

1  using System.Collections.Generic;
2  using System.Collections.Generic;
3  using UnityEngine;
4
5  0 references
6  public class SnowBall : MonoBehaviour
7  {
8      1 reference
9      public float ballSpeed;
10
11      2 references
12      private Rigidbody2D theRB;
13
14      1 reference
15      public GameObject snowBallEffect;
16
17      // Start is called before the first frame update
18      0 references
19      void Start()
20      {
21          theRB = GetComponent<Rigidbody2D>();
22      }
23
24      // Update is called once per frame
25      0 references
26      void Update()
27      {
28          theRB.velocity = new Vector2(ballSpeed * transform.localScale.x, 0);
29      }
30
31      0 references
32      void OnTriggerEnter2D(Collider2D other)
33      {
34          if(other.tag == "Player 1")
35          {
36              FindObjectOfType<GM>().HurtP1();
37          }
38          if(other.tag == "Player 2")
39          {
40              FindObjectOfType<GM>().HurtP2();
41          }
42
43          Instantiate(snowBallEffect, transform.position, transform.rotation);
44          Destroy(gameObject);
45      }
46  }

```

This session worked on my snowball code and physics, similarly to how the player controller used the rigid body to find the velocity. And for the trigger events used the collider 2d to identify and differentiate tags and cause an event in the game managing script.

14

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

0 references
public class DestroyOverTime : MonoBehaviour
{
    2 references
    public float lifeTime;

    // Update is called once per frame
    0 references
    void Update()
    {
        lifeTime -= Time.deltaTime;

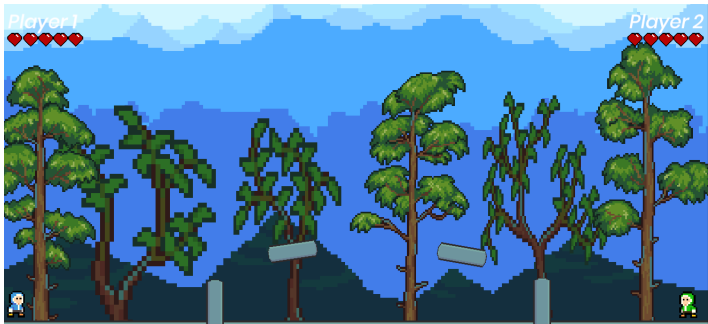
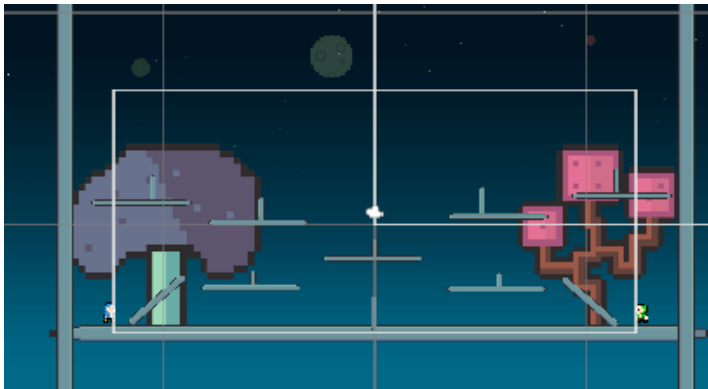
        if (lifeTime < 0)
        {
            Destroy(gameObject);
        }
    }
}
```

I noticed that when the snowballs hit an object the object would not despawn and would bug and make the game lag. It was a simple code making the object be destroyed over time.

Strand C4: Fully justify changes made to the chosen design and plan when making the solution.

List and explain any changes you made to your final design (from B4) while creating your product.

- You can put screenshots to show the comparisons

Change made to Design	Reason
Maps 1  2  3 	The changes made in the design is the layout of the app design. I changed how it looks because believe it aesthetically looks more appealing in comparison to the designs. The designs made in b4 and b2 for the third design were mere placeholders that set the themes not specific layouts. However, for my 3rd map,Ichose to go for a haunted abandoned town design. The reasons for it looking like this is because it matched the theme and was not too distracting but also provided ambience. The structure of the map and the locations of the platforms were placed strategically to prevent common and easy kills allowing fast-paced gameplay as well as a competitive atmosphere by getting rid of options such as “spawn Peaks' ' . (Spawn Peaking is the action of peaking or having the opportunity to kill the opposition seconds after spawning or the match starting) Which in turn allows for a more competitive and challenging environment.
Music	Although I still have a volume slider for it, I decided that music was not a good choice for my product. My product is a fast paced and highly competitive game between 2 people on a single input. The music would have believed to be disruptive to the competitive environment.

List and explain any changes you made to your time and resource plan (from C1) while creating your product due to the changes made to your design

Change made to time plan	Reason
--------------------------	--------

Order of the process, I designed and coded the main menu and information page scene after locating and downloading the necessary assets and a brief plan. Then I designed and coded map 1 then the sprites then map 2 and 3.	The reason for both of these changes is that after planning and doing the designing and coding process. I realized that I have to design and code the structure and main parts of the game first such as things like the main menu, information page, scene loaders and making the sprite be able to do stuff. I appropriately spread my time throughout this project to allow me to balance the act of writing the report, do homework, make the game and balance work life relationships.
The dates in the time plan, I planned and designed my main menu, information page before the new year. After the new year I coded and designed the maps and sprites between late Jan and early Feb.	

Strand D1: Design detailed and relevant testing methods, which generate data, to measure the success of the solution.

Test	Describe your testing method	I will know if the test is successful if...	Success criteria this will check
1	Self Test, I will compare my design specifications with my product.	At maximum, no more than 2 design specifications are lacking or missing as well as if someone is deeply unsatisfied with the product and if it can't solve their problem	all
2	Client trial and observation, I will create a form for my client asked them to answer questions connected with and without my designs specifications.		
3	Audience trial and evaluation, using the same form for my clients will ask a possible client to answer it.		

Strand D2: Critically evaluate the success of the solution against the design specification.

- When explaining the design specification, use the data generated in D1 from self test, client test, target audience survey

Design Specification	Yes/No	Explain the success or partial success or failure
Aesthetics	Yes	The design specification was successful, the product was constructed of an organized, simple and instructive layout that is user-friendly, making navigation and understanding simple and fast. Furthermore, it utilizes modern design styles to capture and retain the attention of the audience by utilizing both light and dark colors. Finally, the use of diverse themes allows for a unique and personalized experience that allows for repetition engagement in coordination with readable fonts and style for the audience.

Function	Yes	I implemented ways to counteract repetitiveness, a health system and made the product competitive. To implore a competitive environment I made winning conditions, health systems and a fast-paced game including set speed and map placement and design. Furthermore, to prevent repetitiveness I added three theme choices for three different maps. Further down the line I will add features to implore multiple maps shuffling for a set theme.
Customer	Yes	Somewhat successful. It works on the newest gen (Gen A, client 2), Gen z (client 1) however have failed to interview a person from the millennial generation.
Development	Yes	I was able to use the internet as a resource to plan, develop and create a product personalized for the client.
Technology	Yes	The client will require internet as well as a web browser to access it, it will be available on every web browser possible to ensure that everyone can access and can download the game from the website, and played on mac, windows and even linux.
Quality	Yes	A simple reboot fixes all the problems
Size	yes	The app covers both the entire screen or in a minimized view. The sprites used have to be a minimum of 1% of the screen to be able to be recognized and the buttons that should be interacted with should not be too big or too small resulting in a minimal size of 4% of the screen. Finally the main text should be a maximum of 8% of the screen.

Strand D3: Explain how the solution could be improved.

- You should use the evaluation done by the client and target audience survey

Improvement	Reason	How will you achieve it
Sprite size,	My client has depicted that the sprite size is small and very hard to track, Thus the fast game feels confusing and overwhelming.	To achieve this I will increase the size of the sprite, however to balance the game mechanics will decrease the hit box to only torso, this allows for a more challenging and competitive game. Furthermore, I can implement an option to choose smaller maps or larger maps, larger maps will keep the small sprites and large maps for those who wish as well as a small map for larger sprites and smaller faster paced maps.
Developed background	My client suggests that the background is basic in the maps. Some layer development is required for more complexity and depth of integration.	I will either design or use some images to create a or multiple backgrounds. A way to further develop this is a scene changing event where either after a set time the scene changes or after winning the next map on the same theme.
Cross-platform multiplayer functionality	My client proposes that implement a cross-platform	I can achieve this by putting this game in a domain and server so people can access and play together and receive inputs separately.

	Multiplayer feature to play online and local games.	
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Strand D4: Explain the impact of the solution on the client or target audience.

- Explain the problem the client and community faced; and the solution that was planned .

The problem the client and community faced was the severe lack of 2 player input games in the industry. Furthermore, the severe lack of games allows for niches to be formed developing repetitiveness between the games and decline of fun and interest in the industry, thus resulting in a lower player count. My product made as a solution to this issue invokes larger interest into this industry for being unique. The unique part is that it invokes competitive emotions and an assortment of themes to increase replayability. The competitive aspects of the game give rise to senses of belonging and promote synergy and working together within a community. Creating routes and sharing that as well as feedback, bus and developmental issues and opinions can spark the rise of an inclusive community.

- Does the product you designed solve the problem you started with? How?

I believe that to an extent it does, the problem discusses the lack of unique 2 player input experiences, although competitive style games do exist my game puts a unique twist. From getting inspiration for map and player movement by games like smash bros and sticking to projectile based action allows fast-paced competitive atmosphere on a single input device. My product allows my client to play a game with their friend on the same keyboard, the game includes a large variety of choice in terms of themes and will get consistent updates in order for those who play the game to get new content such as maps, sprites, and their features.

- How does the product you designed change things for your client and target audience (positive and negative) ? (Also Refer to client evaluation data)

To guarantee an enriching gaming experience my platform offers seamless local multiplayer functionality, allowing two players to enjoy the game without any glitches or technical inconveniences. In the rare event of encountering an error or bug, a straightforward solution is at hand, a simple reboot of the game takes care of the issue swiftly returning players to their gaming adventures. Variation is a cornerstone of the gaming experience, with three distinct themes to choose from including themes such as space, town and even a forest map. Each theme unveils a plethora of different features, each presenting its own set of challenges and strategic opportunities. Players are invited to explore a variety of gameplay scenarios, ensuring that no two gaming sessions are alike. The game's accessibility is of the utmost importance, and the game values itself on its simplicity and user-friendly interface. With user-friendly controls and easy-to-understand mechanics, players of all skill levels can quickly grasp the fundamentals and dive into the action without hesitation. Whether you're a seasoned gamer or a newcomer to the scene, the learning curve is gentle allowing everyone to join in on the fun. Continuous improvement is the aim of the game, with updates occurring regularly, including tweaks to enhance the overall gaming experience. From refining gameplay mechanics to revamping the visual aesthetics. Every aspect of the game receives careful attention to ensure that it remains fresh, engaging, and balanced.

APPENDIX A - Client interaction

Interviewer:	Have you ever played old 2d platformer games	
Client:	Yes I have, I have played pac man before and I enjoyed it a lot. The simple mechanics and the thrill of running away from the ghosts whilst collecting coins is certainly very fun. Another 2D game I play and enjoy a lot is Clash Royale, it is very fun to play with friends and I in general love multiplayer games.	
Interviewer:	What do you think of 2 player games?	
Client:	I love two player games. It is very fun playing with another person, the competitiveness is very entertaining and enjoyable. I always have a very good time playing 2 player games and they are my favorite kind of game.	
Interviewer:	Do you have any previous experience with 2 player games?	
Client:	Yes I do. One of my favorite games as mentioned previously is clash royale because it is a 2D game with a 2 player functionality that just makes it 10X more fun.	
Interviewer:	What type of 2 player games have you played and with whom? LAN, local or global?	
Client:	have played it takes two and Fortnite Duos, as well as previously mentioned clash royale. I also really like local 2 player games.	
Interviewer:	What genres or themes do you resonate with whilst playing 2 player games? Genres can include platformer, space, action etc.)	
Client:	I really like platformer games and especially ones that require some form of shooting to win. These types of games are by far the best for me personally as they fit in perfectly with my extremely competitive personality.	
Interviewer:	What special features do you have that should be implemented in the game?	
Client:	A shooting feature would be crucial for me because I like the feeling of needing to be skillful inorder to win. Another feature is high mobility, I would like the character to be able to move up and down, side to side. I think that that is a very good way to make the game.	
Interviewer:	Do you have any music preferences?	
Client:	I do not have anything particular in mind, only that the song needs to be exciting and thrilling to set the adrenaline mindset for the game.	
Interviewer:	Have you ever played a 2 player game with both players playing on a single keyboard? If so, present examples and thoughts.	
Client:	One of my favorite 2 player games is crossy road and it is played by both of the players being on the same device. I really enjoy this because I get to witness in real time what my opponent is thinking and how they are reacting to the game. It is a lot more fun!	

APPENDIX B - Client Review of Product

Interviewer:	Can you provide comments on the design and construction of my product? Provide details regarding the aesthetics, readability, visuals and how easy it is to understand and navigate.	
Client:	Client 1: I think the game looks incredible. The main menu is very good, the faded background provides a certain depth to the game, the buttons have a very good outline and they are very detailed. Perhaps the title could be slightly bigger, overall the main menu is very good. The design is very easy to understand, and the instructions provided are very clear and useful. When it comes to the map the themes and trees in the background are matching with the sprites, it all follows the same pixel theme and that is very good. It could be even better if the obstacles were themed and if the	Target Audience: I think the game is well-rounded, it provides a range of adequate visuals that make the game easy to navigate and an appropriate amount of words. Furthermore, the use of colors are good but maybe if the gradients were different as well as the colors so that they match a certain theme. Finally think the map layouts are good, the background looks simple but not too distraction which is good and the map designs are unique providing different play styles per map. I would like maybe more maps and the obstacles being relevant to the theme or maybe more obstacles like sprites and such

	sprites were bigger.	
Interviewer:	What is your opinion of the color palette and theme choices available?	
Client:	Client 1: I really like the color pallet, it is very aesthetically pleasing. The purple and blue go very well together in the menu. This same color pallet carries out on the game, and it connects very well.	Target Audience: I like the range of colors and the themes choices however think that more maps with different backgrounds and map designs would be nice.
Interviewer:	Functionally, do you believe the game is competitive and unique as well as not being repetitive. Explain and provide examples	
Client:	Client 1: Yes the game is very competitive especially when the winner is announced, this provides greater competition amongst me and client 2 which makes it very fun. I enjoy the game very much because it is competitive and a borderline shooter.	Target Audience: Yes, I compare my friends and myself when playing this game and seeing who is better, however I think that it would be better to make a multiplayer and LAN feature available.
Interviewer:	Is the game easy to learn and play?	
Client:	Client 1: The game is very easy to learn because there is a set of instructions when you press play that tells you exactly which keys to use. It is not confusing at all, everything is very clear and I did not need to contact the developer at any point to ask how the game works.	Target Audience: Yes, the controls are visually depicted, allowing easy and fast comprehension. Furthermore due to the simple and easy to pick up controls the gameplay is also easy to learn.
Interviewer:	Is the game easily able to support simple navigation?	
Client:	Client 1: Yes, the game certainly follows simple mechanics, which provides a very simple and easy to use navigation.	Target Audience: yes the buttons and details provided allow easy and fast navigation.
Interviewer:	Is the game fun and why?	
Client:	Client 1: The game is very fun because it provides a fun multiplayer experience that is perfect for people as competitive as me. The game would be even better if it was accessible for players across platforms and it can be connected remotely.	Target Audience: yes, the game is primarily fun due to the different maps having different map layouts, allowing different play styles to be planned and implemented.
Interviewer:	Is the game appropriate for the target audience?	
Client:	Client 1: The game is very suitable for anyone who enjoys 2D laptop games, as well as multiplayer local functionalities. The age range is very suitable for people in the generation Z and millennials.	Target Audience: Yes, it is both fun and appropriate. It is A perfect game for simple and fast leisure to those on Laptop to PC at home and even on plane traveling.
Interviewer:	Does it meet both your expectations and is there any feedback?	
Client:	Client 1: The game certainly meets my expectations, however I do wish that it was accessible for cross-platform players because me and client 2 live very far away from each other so it is hard for us to find the time to meet up just to play the game.	Target Audience: Yes, I personally think that adding obstacles like traps, spikes and such would add a difficulty feature. More maps would be nice and even a customizable choice of the projectiles and sprites.
Interviewer:	DO the animations, sprites and other designs satisfy your needs?	
Client:	Client 1: Overall I feel very satisfied with the	Target Audience: Yes, in terms of the animation and

	animations, especially the moving left and right animations are very clean and I like them very much. One issue I have is that I wish the sprites themselves were slightly bigger and easier to see, I would also prefer if the background was more detailed.	sprites I'm satisfied,I would like as previously stated more maps and obstacles and a customizable feature
Interviewer:	Are you satisfied with most platform accessibility?	
Client:	Client 1: I really enjoy the accessibility of a local multiplayer game, it is fun because I can socialize face to face with client 2 whilst having fun in a game.	Target Audience: yes,I play on both PC and laptop it works and runs smoothly, however I would like a mobile version.
Interviewer:	Does it solve your problem? Why?	
Client:	Client 1: The game definitely solves my problem, it provides me with a fun and interactive multiplayer platformer game. I had previously struggled to find such a game available for download for my iMac, however now I feel as if all my needs have been met and my problem has been solved.	Target Audience: Definitely, it provides a unique two player input experience that is fun and allows the user to choose and transition between themes.
Interviewer:	Are you satisfied with the game overall?	
Client:	Client 1: Overall the game is very good. The multiplayer functionality is great and works perfectly, all the animations are on point, the color scheme is very good and in general it solves my issues very well. A small thing I would change would be to increase the sprint size, more complex background and perhaps if time allows a cross platform multiplayer functionality.	Target Audience: yes.
Interviewer:	Would you recommend this game to others	
Client:	Client 1: I would certainly recommend this game for any millennial or Gen Z who enjoy playing competitively 2D video games together.	Target Audience: Definitely.

APPENDIX C - Answers to Research

Nº	Source	Usage
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1

<https://ben-pielstick.medium.com/blinking-text-in-unity-1dd28365a476>

Sadly(?) Unity doesn't support HTML blink tags. Actually this might be for the best, but if you're really interested in making blinking text, it isn't too hard actually.

```
private IEnumerator GameOverBlink()
{
    while (!_gameover)
    {
        if (_gameOverText.activeInHierarchy)
        {
            _gameOverText.SetActive(false);
        } else
        {
            _gameOverText.SetActive(true);
        }
        yield return new WaitForSeconds(0.5f);
    }
    yield break;
}
```

All it takes is StartCoroutine(GameOverBlink()); to get this going

Coroutines are the basic way to do pretty much anything with timing in Unity. In this case all that's happening is flipping the active state of a game object off and on every 0.5 seconds, but coroutines can do any number of other useful things. Of course we could also turn toggle the same state on the text component of the game object, or even blank out the text and fill it in again, but accessing game objects at the top level is the easiest.

From this source I used the wait for, enumerator and yield return new, in summary I used this source to allow my code to be used during multiple frames and have a wait time in between and stopping the process.

2

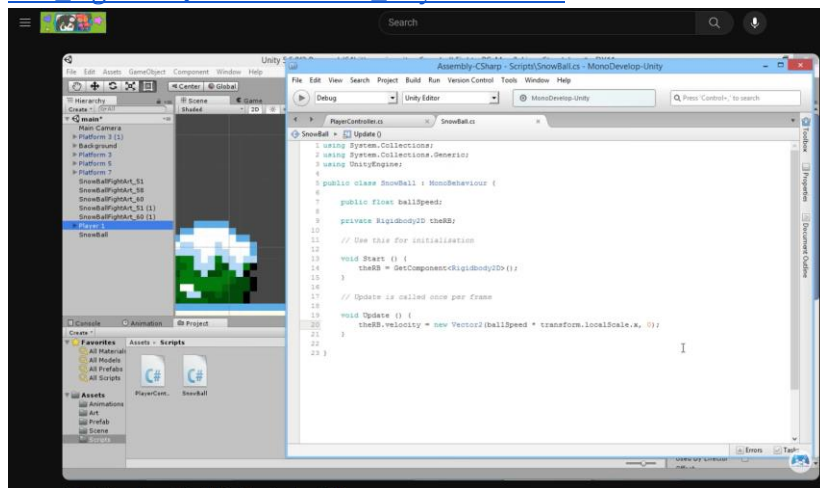
<https://www.youtube.com/watch?v=JeZkctmoBPw>



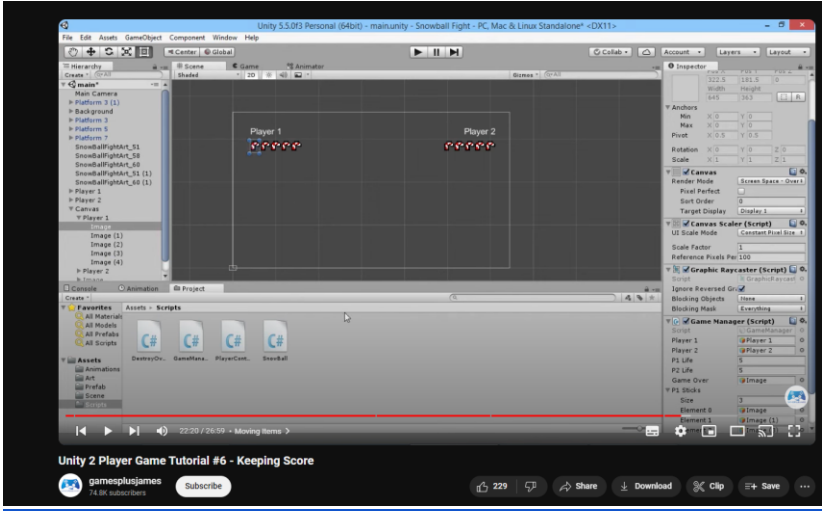
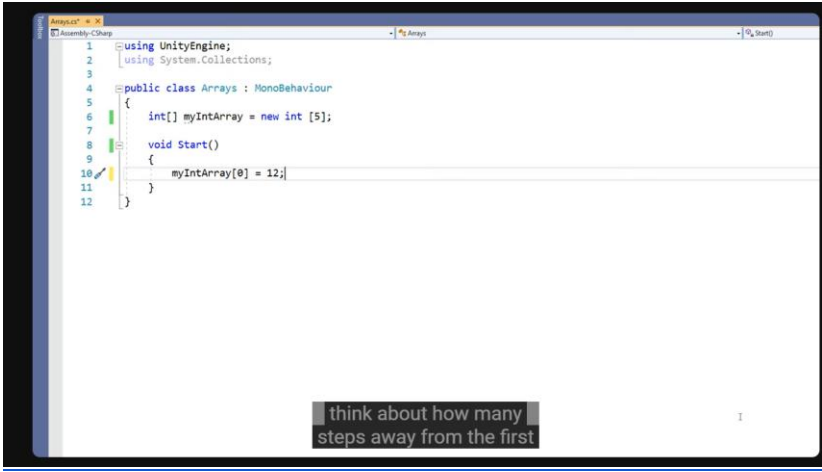
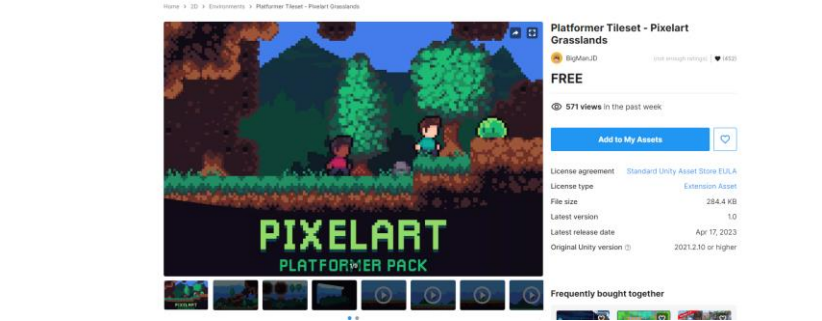
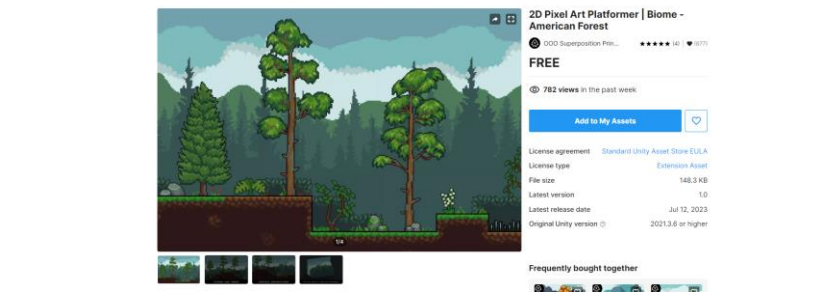
I used this to learn how the animator works

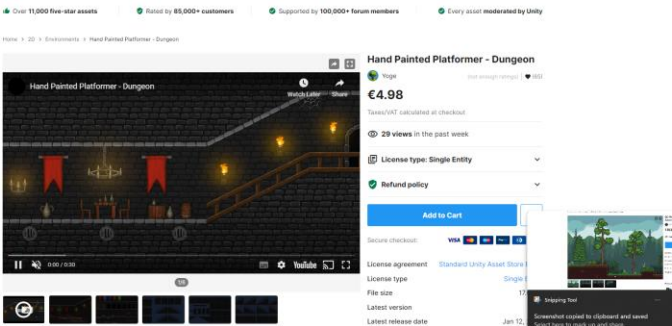
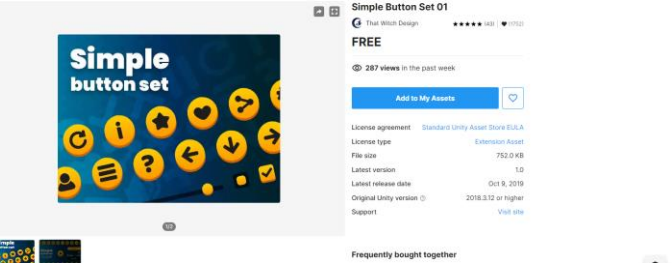
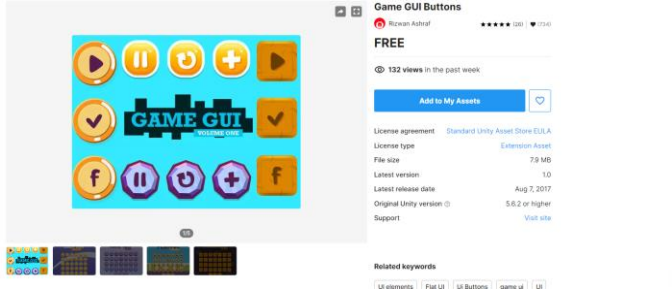
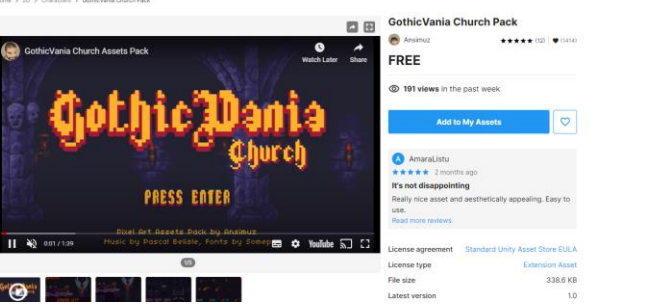
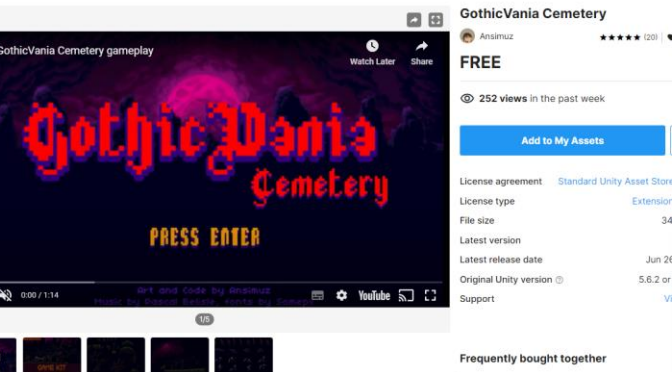
3

https://www.youtube.com/watch?v=xfvsUhNQLtA&list=PLiyfvmjtjWC_Ugm9c9Q7WaoRFGbZh_Z6ys&index=6



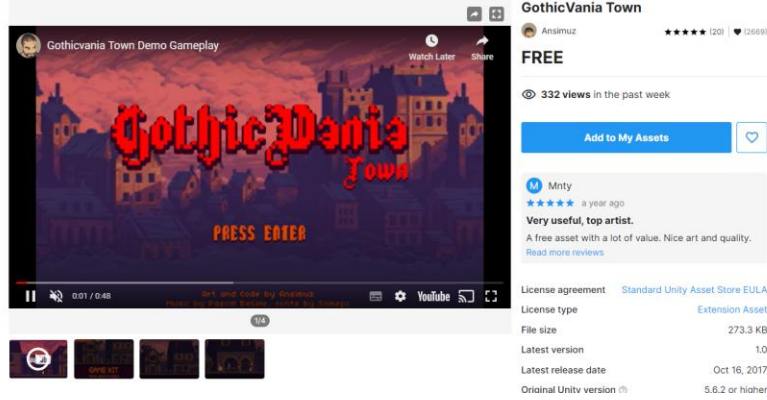
I used this to start the health array

4	https://www.youtube.com/watch?v=Zn4BDIXhy-M 	I used this source to learn about arrays and combine them with my health systems and tag knowledge
5	https://www.youtube.com/watch?v=lwzUiOCOqVA&list=PLiyfvmtjWC_Ugm9c9Q7WaoRFGBZh_Z6ys&index=4  <pre> 1 using UnityEngine; 2 using System.Collections; 3 4 public class Arrays : MonoBehaviour 5 { 6 int[] myIntArray = new int [5]; 7 8 void Start() 9 { 10 myIntArray[0] = 12; 11 } 12 } </pre> think about how many steps away from the first	I used this to learn about how to find the ball speed in a projectile and how to destroy the recurring game object
6	https://www.dropbox.com/s/s85h2o5pamutj0p/Snowball_Fight_Reso-urces.zip?dl=0	Sprites for main player 1 and 2 and snowball and animations
7	https://assetstore.unity.com/packages/2d/environments/platformer-tileset-pixelart-grasslands-248158 	Forest background
8	https://assetstore.unity.com/packages/2d/environments/2d-pixel-art-platformer-biome-american-forest-255694 	Forest trees

9	https://assetstore.unity.com/packages/2d/environments/hand-painted-platformer-dungeon-240848 	Dungeon stairs and platforms
10	https://assetstore.unity.com/packages/2d/gui/icons/simple-button-set-01-153979 	Settings buttons
11	https://assetstore.unity.com/packages/2d/gui/icons/game-gui-buttons-96277 	Most used buttons
12	https://assetstore.unity.com/packages/2d/characters/gothicvania-church-pack-147117 	Town background and map
13	https://assetstore.unity.com/packages/2d/characters/gothicvania-cemetery-120509 	

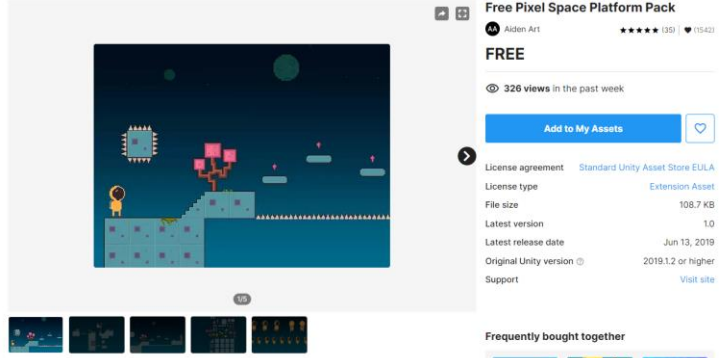
14

<https://assetstore.unity.com/packages/2d/characters/gothicvania-town-101407>



15

<https://assetstore.unity.com/packages/2d/characters/free-pixel-space-platform-pack-146318>



Space map

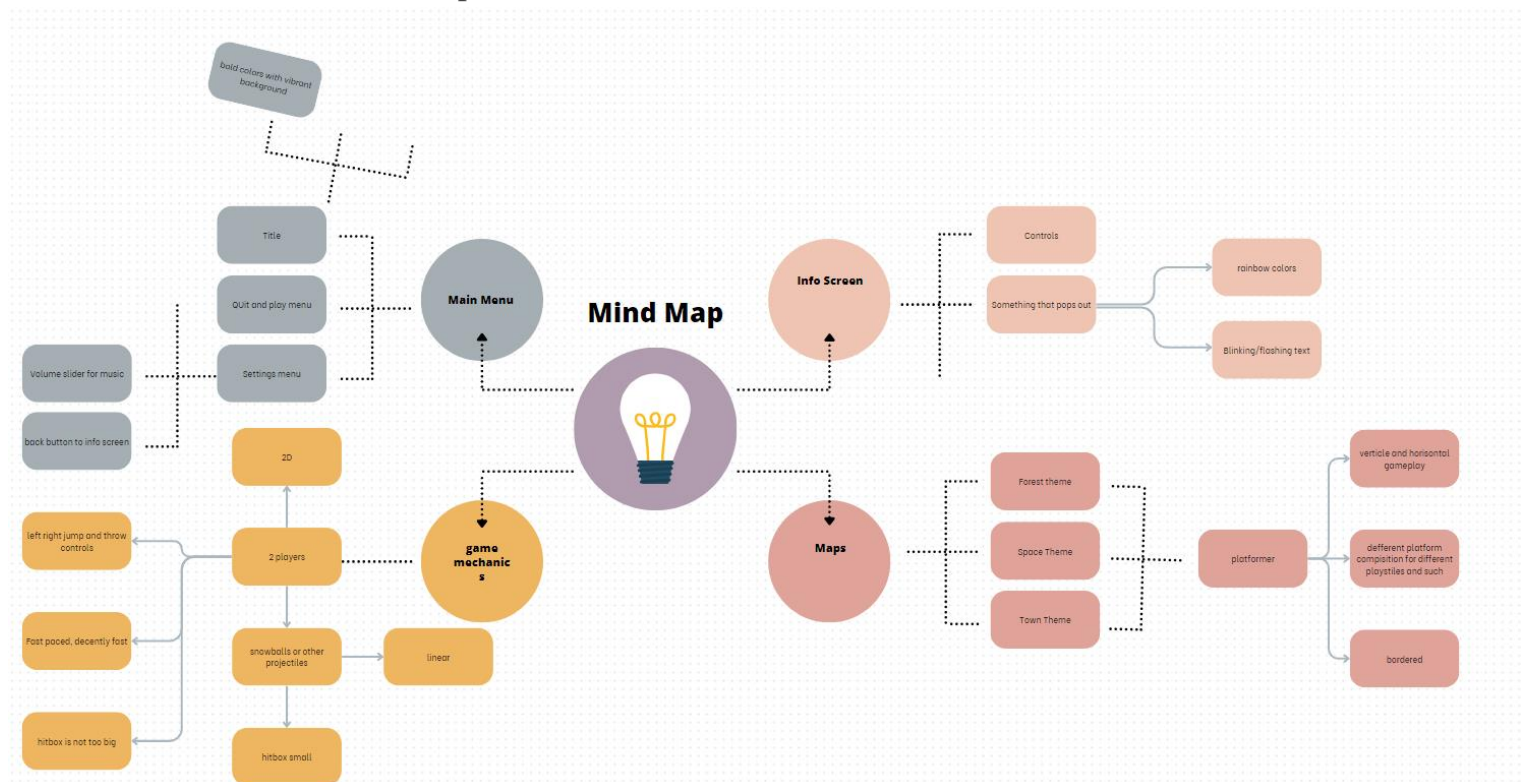
APPENDIX D - Designs

<https://app.moqups.com/lsMiDcPSQTn0v3xqBdCF9OSfQyNnC02D/view/page/ad64222d5?ui=0>

APPENDIX E - Video

Video

APPENDIX F - Plan/Mindmap



APPENDIX G - Bibliography

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